Arsenic trioxide unmasks selective suppression of nucleotide excision repair in African American breast cancer stem-like cells: a transcriptome-wide study

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ABSTRACT

African American women experience approximately 40% higher breast cancer mortality than non-Hispanic White women despite similar incidence rates. This disparity is driven by incompletely characterized biological and environmental factors. Arsenic, an established environmental carcinogen with evidence for racial variation in exposure, is a known inhibitor of nucleotide excision repair (NER), but whether this inhibition differs by racial background or is amplified in the breast tissue-derived cancer stem cell compartment has not been investigated. Here we report transcriptome-wide profiling of two breast tissue-derived cell lines representing African American (BRL-24) and non-Hispanic White (BRL-23) racial backgrounds across parental and CD24−/CD49f+/CD44+ sorted stem-like cancer stem cell-enriched populations under vehicle and chronic low-dose arsenic trioxide conditions (0.3 μM, 14 days, approximately the LC10). CD24−/CD49f+/CD44+ sorted cells were isolated by flow cytometric sorting to enrich for a CSC-like population. CD44 and CD24−/low identify tumor-initiating cells in breast cancer, CD49f (α6-integrin) marks mammary epithelial progenitors with self-renewal capacity, and the combined CD24−/CD49f+/CD44+ panel has been validated to isolate subpopulations with self-renewal, tumor-initiating, and elevated DNA repair properties across breast cancer and DCIS model systems. RNA from all 16 samples was profiled by paired-end RNA sequencing and analyzed using DESeq2 with a full factorial interaction model followed by GSEAPreranked across four MSigDB databases (48 total runs; significance: |NES| > 1.5, FDR < 0.25). Sorted populations were validated as genuine CSC-like populations by enrichment of published embryonic and mammary stem cell gene signatures and coordinate elevation of BER, MMR, and HR pathway programs. While arsenic collapsed 68% of baseline racial transcriptional differences in parental cells, racial differences were amplified approximately tenfold in the stem cell compartment (582 versus 71 significantly enriched pathways). The primary finding was selective suppression of KEGG_NUCLEOTIDE_EXCISION_REPAIR specifically in African American stem cells under arsenic (NES = −2.75, FDR = 0.18), confirmed independently by Reactome (NES = −2.68), affecting both the damage recognition stage (XPC, DDB1, CETN2, RAD23A, ERCC3, GTF2H1) and the gap-filling and ligation machinery (PCNA, LIG1, RFC3–5). No equivalent suppression was observed in non-Hispanic White stem cells under arsenic (NES = +0.86) nor in parental cells of either line under any condition, establishing that the racial NER disparity requires the simultaneous combination of African American cellular biology, CSC-like identity, and arsenic exposure. ERCC1, the obligate NER incision endonuclease subunit, was consistently suppressed in African American stem cells across four comparisons and present in both independent racial NER leading edges, identifying it as the primary candidate for functional validation. These findings provide the first transcriptomics-level mechanistic framework connecting racial background, arsenic exposure, CSC-like biology, and impaired genomic maintenance relevant to breast cancer health disparities.

Keywords: *breast cancer; racial disparities; arsenic trioxide; nucleotide excision repair; ERCC1; cancer stem-like cells (CSC-like); CD24−/CD49f+/CD44+; RNA sequencing; gene set enrichment analysis; health disparities*

Interactive Data Visualization Site: https://aubfigz.github.io/Arsenic-Project/

INTRODUCTION

Breast cancer is the most frequently diagnosed cancer and the second leading cause of cancer death among women in the United States.[28] Significant racial disparities persist in breast cancer outcomes: African American (AA) women experience approximately 40% higher breast cancer mortality than non-Hispanic White (NHW) women and are more likely to be diagnosed with aggressive, high-grade tumors at younger ages.[29,30] Environmental exposures including endocrine-disrupting chemicals and heavy metals are increasingly recognized as contributors to these disparities, yet the specific mechanisms by which racial background modifies vulnerability to environmental carcinogens remain incompletely characterized.[31]

Arsenic is a Group 1 human carcinogen classified by the International Agency for Research on Cancer and an established environmental carcinogen with documented associations with breast cancer risk, particularly for premenopausal women and with evidence for racial variation in tissue-level sensitivity.[32] In the United States, arsenic contamination of drinking water affects millions of people, with particularly high levels documented in communities across South Florida and other Southern states that also carry elevated rates of advanced cancer diagnosis.[4] Arsenic promotes carcinogenesis through multiple mechanisms including oxidative stress, epigenetic dysregulation, and - critically - inhibition of DNA repair.[33,34] Unlike classical mutagens that directly cause point mutations, arsenic acts primarily as a co-carcinogen by impairing the capacity of cells to remove DNA damage caused by other environmental insults, amplifying their mutagenic effect.[35]

Nucleotide excision repair (NER) is the primary DNA repair pathway responsible for removing a broad spectrum of helix-distorting lesions, including bulky adducts induced by polycyclic aromatic hydrocarbons, UV radiation, and arsenic-associated oxidative damage.[36,37] NER operates through two sub-pathways: global-genome NER (GG-NER), which surveys the entire genome for lesions, and transcription-coupled NER (TC-NER), which preferentially repairs damage in actively transcribed genes. Both converge on a common incision machinery involving the ERCC1-XPF endonuclease complex, which executes the critical 5′ incision step.[41,42] Arsenic inhibits NER through multiple mechanisms: it reduces expression of the damage recognition factor XPC, displaces zinc from the zinc finger domains of NER proteins XPA and PARP-1, and suppresses p53-mediated transcriptional activation of NER genes.[35,36,37] The consequence is that cells co-exposed to arsenic and other carcinogens accumulate unrepaired NER substrates at a higher rate, potentiating carcinogenesis. Importantly, prior work characterizing arsenic-NER interactions has been conducted in race-indeterminate bulk cell populations; whether the magnitude or context-dependence of arsenic-induced NER impairment differs by racial background has not been investigated.

Cancer stem-like cells (CSC-like cells) — operationally defined by surface marker panels that identify self-renewing, progenitor-enriched subpopulations — are implicated in breast cancer initiation, therapy resistance, and recurrence.[6] In breast tissue-derived cell systems, such CSC-like populations are identified by the surface phenotype CD24−/CD49f+/CD44+: loss of the differentiation marker CD24, retention of the mammary progenitor marker CD49f (integrin α6), and co-expression of the adhesion receptor CD44.[6,7] Al-Hajj et al. established that CD44+/CD24−/low marks tumor-initiating cells in primary human breast cancer,[6] and Shackleton et al. demonstrated that CD49f (α6-integrin) enriches for mammary epithelial progenitors with self-renewal and multipotent capacity from normal mammary tissue.[7] Together, the CD24−/CD49f+/CD44+ panel isolates a progenitor-enriched subpopulation with enhanced self-renewal capacity, elevated DNA repair programs, and resistance to genotoxic stress. Importantly, the functional properties conferred by this panel — self-renewal, tumor-initiating capacity, and elevated DNA repair — have been independently validated in human DCIS and breast cancer cell line models.[8] Accordingly, populations sorted by this panel in the present study are designated “CSC-like” to reflect their functional definition by established stem cell markers, while acknowledging that BRL-23 and BRL-24 originate from non-diseased mammoplasty tissue rather than from breast tumors.[8]

This study addresses the intersection of these three underexplored areas - racial biology, arsenic-induced DNA repair impairment, and cancer stem cell transcriptomics - using a genome-wide RNA sequencing approach with a specific design to detect accumulated small changes in NER and DNA repair pathway gene expression rather

than relying on individual gene thresholds. We profiled two breast tissue-derived cell lines representing AA (BRL-24) and NHW (BRL-23) racial backgrounds across parental and CD24−/CD49f+/CD44+ sorted CSC-like populations, under vehicle and chronic low-dose arsenic trioxide conditions, generating 12 pairwise contrasts. Pathway-level analysis using Gene Set Enrichment Analysis enabled detection of coordinated, sub-threshold changes across NER gene sets and confirmed stem cell population identity through enrichment of published cancer stem cell (CSC-like) signatures. The central finding - that arsenic unmasks a selective suppression of NER specifically in AA CSC-like cells that is absent in NHW stem cells and absent in parental cells of either line - identifies the CSC-like stem cell compartment as the primary locus of racial biological divergence relevant to arsenic-associated breast cancer risk.

MATERIALS AND METHODS

1. Cell Lines and Maintenance

Two patient-derived breast tissue cell lines were used in this study: BRL-23 (NHW, 22-year-old premenopausal donor) and BRL-24 (AA, 33-year-old premenopausal donor), derived from non-diseased breast reduction mammoplasty tissue obtained from premenopausal women under institutional review board approval (UPMC IRB 0609002) and characterized in prior work from this laboratory.[1,3] BRL-23 and BRL-24 were selected as the pair of non-diseased mammoplasty-derived lines in the Latimer laboratory collection that most closely match on donor age range, menopausal status, and karyotype integrity while differing in self-reported racial background (NHW and AA, respectively), making them the most appropriate matched pair available for an unconfounded racial comparison. Both parental cell lines were originally established and karyotyped in the Latimer laboratory and were subsequently used in the arsenic dose-response characterization and prior arsenic-exposure transcriptomic studies reported by Alharbi and Yousef. Karyotyping confirmed normal chromosomal content for both lines. Cells were cultured in MWRI medium as originally formulated by the Magee-Womens Research Institute and previously described by Yousef (2025) and Alharbi (2025),[1,3] composed of DMEM (HyClone, Cat. SH30243.01) supplemented with 10% heat-inactivated fetal bovine serum (HyClone, Cat. SH30084.03), 10% heat-inactivated newborn calf serum (HyClone, Cat. SH30118.04), 1% penicillin/streptomycin (Corning, Cat. 30-002-CI), 1% MEM non-essential amino acids (Corning, Ref. 25-025-CI), 1% beta-mercaptoethanol (Sigma-Aldrich, Cat. M3148), and 1% nucleoside stock (Sigma-Aldrich), filter-sterilized through a 0.45 µm membrane. Cells were maintained in 12.5 cm^2 flasks pre-coated with a 1:1 dilution of Matrigel (Corning) and DMEM at 37°C with 10% CO_2. The flow-cytometric enrichment of CD24−/CD49f+/CD44+ CSC-like populations described below (Section 3) was performed on these parental cultures in the present study; prior work from this laboratory (Yousef, 2025; Alharbi, 2025) reported arsenic exposure and transcriptomic analyses on the unsorted parental BRL-23 and BRL-24 lines only, and did not include the CSC-like-sorted populations.

2. Arsenic Trioxide Exposure and Experimental Design

The arsenic trioxide (ATO; arsenic(III) oxide, CAS 1327-53-3; Sigma-Aldrich) dose was established through a cell viability assay performed in the parental BRL-23 and BRL-24 lines as described by Alharbi (2025) and Yousef (2025),[3,1] identifying the LC10 as 0.3 µM, consistent with arsenic concentrations in contaminated groundwater.[3,4] For the transcriptomic experiment, cells were seeded, allowed to attach for 24 hours, then divided into vehicle (PBS) and arsenic-treated groups. A 40 mM ATO stock solution was prepared in PBS, filter-sterilized (0.22 µm), and stored at 4°C protected from light, following the stock-preparation and storage protocol previously established in this laboratory[1,3]. Working solutions were prepared fresh before each feeding. Arsenic-treated cells received 0.3 µM ATO; vehicle controls received equivalent PBS volume. Both groups were fed every other day plus one hour before harvest for 14 days (approximately two to three population doublings), modeling chronic environmental arsenic exposure.[3,5]

At the end of the 14-day exposure, cells were harvested for flow cytometric isolation of CSC-like-enriched populations, producing four biological groups per cell line: Parental Vehicle (PV), Parental Arsenic (PA), Stem Vehicle (SV), and Stem Arsenic (SA). Parental conditions were collected at n=3 biological replicates per group per cell line; stem cell conditions yielded n=1 per group due to sorting yield constraints. The sample inventory is summarized in Table 1.

Table 1. Sample inventory for RNA sequencing analysis.

Cell Line	Condition	Population	n	Comparison IDs
BRL-23 (NHW)	Vehicle	Parental	3	C1, C2, C3, C5, C7
BRL-23 (NHW)	Arsenic (0.3 µM)	Parental	3	C2, C3, C7
BRL-23 (NHW)	Vehicle	Stem (CD24−/CD49f+/CD44+)	1	C5, C9, C11
BRL-23 (NHW)	Arsenic (0.3 µM)	Stem (CD24−/CD49f+/CD44+)	1	C7, C9, C12
BRL-24 (AA)	Vehicle	Parental	3	C1, C2, C4, C6, C8
BRL-24 (AA)	Arsenic (0.3 µM)	Parental	3	C2, C4, C8
BRL-24 (AA)	Vehicle	Stem (CD24−/CD49f+/CD44+)	1	C6, C10, C11
BRL-24 (AA)	Arsenic (0.3 µM)	Stem (CD24−/CD49f+/CD44+)	1	C8, C10, C12

Total: 16 samples. AA = African American; NHW = Non-Hispanic White. Stem populations isolated by CD24−/CD49f+/CD44+ surface expression.

3. Cancer Stem Cell Enrichment by Flow Cytometry

CSC-like-enriched populations were isolated from both vehicle- and arsenic-treated cultures at the end of the 14-day treatment period using a three-marker surface phenotype: CD24−/CD49f+/CD44+. Flow-cytometric enrichment of the CSC-like fraction was performed in the present study and was not part of the prior Yousef (2025) and Alharbi (2025) arsenic-exposure experiments, which analyzed unsorted parental BRL-23 and BRL-24 cultures only.[1,3] This panel is established for isolating progenitor-enriched populations from breast tissue-derived cell lines, with CD24− indicating loss of differentiation antigen, CD49f+ (integrin α6) marking mammary epithelial progenitors, and CD44+ identifying the CSC-like stem cell adhesion receptor.[6] Cells were harvested by trypsinization, washed, and resuspended in PBS with 2% FBS for antibody staining. The following antibodies were used: anti-human CD24-FITC (BioLegend), anti-human CD49f-PE (BioLegend), and anti-human CD44-APC (BioLegend), each incubated at 4°C for 20 minutes. Viable cells were identified by DAPI exclusion. Fluorescence minus one (FMO) controls were used to define gating boundaries for each marker. Flow cytometric sorting was performed on a BD FACS Aria instrument; CD24−/CD49f+/CD44+ cells

were collected directly into RNAprotect Cell Reagent (Qiagen) for immediate RNA stabilization. The triple-gated population consistently represented 2-5% of viable cells across both lines and conditions.

4. Total RNA Extraction and Quality Assessment

Total RNA was extracted from all 16 samples using the miRNeasy Mini Kit (Qiagen, Cat. 217004) with on-column DNase digestion (RNase-Free DNase Set, Qiagen, Cat. 79254). Parental cells were lysed with QIAzol directly in the flask, phase-separated with chloroform, and bound to RNeasy Mini spin columns following the manufacturer's protocol. RNA was eluted in 35 µL RNase-free water. Concentration and purity were assessed by NanoDrop Lite spectrophotometry (A260/A280 1.9-2.1 for all samples). RNA integrity was assessed using the Agilent TapeStation 4200 (RNA Screen Tape, Cat. 5067-5576); all 16 samples achieved RIN ≥ 9, confirming high-quality input.

5. RNA Library Preparation and Sequencing

Libraries were prepared using the TruSeq Stranded Total RNA Library Prep Gold kit (Illumina, Cat. 20020598) and sequenced on an Illumina NextSeq 2000 at the Center for Collaborative Research Genomics Core Facility, Nova Southeastern University (NextSeq 500 High Output Kit, 300-cycle; 1200M reads flow cell). All samples were sequenced as 150 bp paired-end reads at 41-72 million read pairs per sample.

6. Pre-Alignment Quality Control and Batch Structure Assessment

Pre-alignment QC was performed in Partek Flow (v11.0.24.0226) using FastQC-equivalent metrics.9 Arsenic-treated parental samples showed average Phred scores of approximately 33 versus approximately 38 for vehicle and stem samples; stem samples showed elevated %N (0.24% vs. 0.01-0.07%), indicating separate library preparation batches. Because batch structure was inseparable from biological treatment groups, standard ComBat9 correction could not be applied without removing biological signal; confounding was instead addressed through the full covariate structure of the DESeq2 model (Section 9).

7. Read Alignment and Gene Quantification

Reads were aligned to GRCh38 (hg38) using STAR11 within Partek Flow against Ensembl release 105 annotation in paired-end mode, a choice supported by systematic benchmarking of spliced alignment programs,[12] achieving 90-92% alignment rates. Gene-level counts were generated using the quantify-to-annotation-model function (equivalent to featureCounts13), requiring strict paired-end compatibility, annotated intron boundary matching for junction reads, and a minimum 75% read overlap threshold. A summed count filter of ≥10 reads across all 16 samples yielded a final matrix of 16,682 genes × 16 samples.

8. Normalization and the DESeq2 Workflow

Raw (unnormalized) gene counts served as input to the DESeq2 analysis pipeline within Partek Flow, as required by DESeq2, which performs its own size factor normalization internally. DESeq2 applies the median-of-ratios method[14] as part of its internal workflow: each gene's pseudo-reference count is the geometric mean across all samples; each sample's size factor is the median of the ratios of its counts to these pseudo-references; counts are divided by size factors before statistical modeling. This corrects for both library size and library composition - a distinction critical for arsenic experiments, where global transcriptional suppression by arsenic could bias simpler approaches. Counts Per Million (CPM) normalization, used in prior work from this laboratory (Alharbi, 20253) for simpler two-group designs, corrects only for sequencing depth by dividing counts by total reads. Because CPM uses the total count sum as its denominator, a few highly expressed genes that dominate the library can inflate the apparent expression of all other genes - a composition bias especially problematic when arsenic globally suppresses transcription. DESeq2-normalized counts produced by this internal workflow were used for all downstream visualization including principal component

analysis. Successful normalization was confirmed by box plot inspection showing identical median heights and interquartile ranges across all 16 samples.

The practical consequence of this composition bias was directly observed when CPM-normalized counts from this dataset were compared to DESeq2 results across the same 16 samples. For within-line arsenic comparisons in parental cells (C3 and C4 equivalents), CPM-derived L2FC values were directionally consistent with DESeq2 results for most genes — for example, SMO (CPM L2FC −0.76 vs. DESeq2 −1.12 in NHW; CPM −1.38 vs. DESeq2 −1.52 in AA) and KLF4 (CPM −1.23 vs. DESeq2 −1.38 in NHW), reflecting that within-line comparisons using the same total read counts in denominator are less susceptible to composition distortion. However, a parallel CPM-based analysis of 54 key genes using LOG(average_group2/average_group1) fold changes revealed severe composition bias in cross-line racial comparisons. ALDH1A3 is the most illustrative example: CPM from the curated-gene analysis reported it as 7.6-fold lower in AA cells (L2FC −2.92), while DESeq2 identified it as 2.9-fold higher (L2FC +1.51) — an opposite-direction error exceeding 4 $\log_2$ units. The mechanism is that VIM, constitutively expressed at ~3,000–4,000 CPM per million in BRL-23 and higher in BRL-24, dominates the total count denominator differently in the two lines and systematically inflates or deflates every other gene's apparent CPM ratio. DESeq2's median-of-ratios normalization, which uses the geometric mean of each gene as its reference rather than the total count sum, corrects for this; CPM does not. When the full normalized count matrices were compared directly for NER genes in the C8-equivalent comparison (AA Stem Arsenic vs. AA Parental Arsenic), CPM and DESeq2 were directionally identical for all 23 NER genes examined — confirming the NER suppression signal is present in both normalizations at the individual gene level, and that the critical failure of CPM is not gene-level direction but its inability to generate the ranked list, statistical framework, and GSEA pathway analysis that reveals the coordinated pathway-level suppression. Additionally, ALDH1A3 showed a direction reversal in the AA arsenic vs. vehicle parental comparison (C4 equivalent): CPM reported +1.45 (induced by arsenic in AA) while DESeq2 reported −1.29 (suppressed) — a biologically significant error that would lead to the wrong conclusion about arsenic's effect on this stem cell marker in AA cells.

9. Differential Expression Analysis: DESeq2 and the Choice Over CPM + Log2FC

Differential expression was performed using DESeq2 version 1.3813 within Partek Flow. The DESeq2 + GSEA analytical pipeline was designed around three specific goals: (1) to detect small but consistent accumulations of expression change across NER and DNA repair pathway genes that individually fall below significance thresholds but collectively represent a coordinated biological response to arsenic; (2) to validate that sorted stem cell populations were genuinely transcriptionally distinct from parental cells; and (3) to identify racial differences between BRL-23 and BRL-24 across all conditions and cell types. The choice of DESeq2 over the simpler CPM + Log2FC approach - used in prior work from this laboratory for simpler two-group designs - was made because CPM provides no statistical testing framework, no mechanism for detecting accumulated pathway-level changes, no accommodation for the factorial interaction model, and no ability to borrow variance across genes for n=1 stem cell conditions. Calculating fold change and a p-value directly from CPM-normalized counts using a t-test or ANOVA also violates distributional assumptions, as RNA-seq counts are overdispersed count data, not normally distributed continuous measurements.

DESeq2 models counts using the negative binomial distribution, appropriate for overdispersed count data.[13] Gene-wise dispersion is estimated through empirical Bayes shrinkage, critical for n=1 stem cell conditions where within-group variance cannot be estimated directly; DESeq2 imputes variance from the cross-gene dispersion relationship established in the n=3 parental conditions, enabling principled analysis of all 12 comparisons.[15] Log2 fold change shrinkage was applied using apeglm,[16] stabilizing estimates for lowly expressed genes and preventing noisy estimates from dominating GSEA ranking.

The statistical model included all main effects and pairwise interactions: ~ ethnicity + Treatment + Type + ethnicity*Treatment* + *ethnicity*Type + Treatment*Type Reference levels: Vehicle (treatment), Parental (cell type), NHW (ethnicity). The ethnicity-by-treatment interaction directly tests whether arsenic affects gene expression differently in AA versus NHW cells; treatment-by-type tests whether arsenic acts differently in stem

versus parental cells; ethnicity-by-type tests whether stemness differs between lines. Twelve pairwise contrasts (C1–C12) were defined from this model, corresponding to the results presented in Figures 1–4; three additional interaction contrasts (ethnicity×treatment, ethnicity×type, treatment×type) were also evaluated from the same model for supplementary inference but are not reported as numbered comparisons. Individual gene significance: FDR < 0.05 (Benjamini-Hochberg). Standard t-tests and ANOVA were not used because both assume normally distributed data, a condition violated by RNA-seq count distributions; neither can accommodate the overdispersion of count data nor the n=1 stem cell conditions. The prior publication from this laboratory (Alharbi, 20253) appropriately used CPM + paired t-tests for a simpler two-group design; the present study requires the more sophisticated pipeline to handle the factorial model, detect accumulated pathway-level changes, and analyze n=1 stem cell conditions.

Two further limitations of the CPM approach were directly observed in the parallel analysis. First, every stem cell comparison sheet in the CPM analysis contained only a single sample per group — one stem vehicle and one stem arsenic sample per line — making t-tests impossible and leaving all stem cell comparisons without p-values or any statistical framework. The C8-equivalent comparison (AA stem vs. AA parental arsenic, the comparison yielding the primary NER finding) was therefore reduced to a bare ratio of two single CPM values with no uncertainty estimate. DESeq2's empirical Bayes dispersion imputation, which borrows variance information from the n=3 parental conditions to enable principled analysis of the n=1 stem cell conditions, is the only mechanism that makes statistically valid inference possible for these comparisons. Second, the curated 54-gene list used in the CPM analysis contained no NER pathway genes — not a single member of KEGG_NUCLEOTIDE_EXCISION_REPAIR (XPC, DDB1, ERCC1, ERCC3, CETN2, RAD23A, RFC3–5, PCNA, LIG1, POLD1, etc.) was included. This is not accidental: individual NER genes change by 1.2- to 1.8-fold under arsenic in the relevant comparisons, changes too small to attract attention in a gene-by-gene CPM review and too subtle to survive 16,682-gene multiple testing correction as individual DEGs. The NER finding depends entirely on GSEA detecting the coordinated clustering of all these sub-threshold genes at the bottom of the ranked list — a signal that is structurally invisible to any gene-list-based CPM analysis regardless of how many genes are included.

10. Gene Set Enrichment Analysis: Rationale, Goals, and Comparison to Alternative Methods

Pathway-level analysis was performed using GSEA version 4.4.0.[17,18] Three specific analytical goals drove the choice of GSEA over alternative methods. Goal 1: Detecting accumulated small pathway changes in NER and DNA repair. DNA repair pathway genes are constitutively expressed housekeeping genes whose biologically meaningful responses to arsenic are often 1.2- to 1.8-fold changes that may not individually exceed FDR correction thresholds. GSEA detects this biology by evaluating whether NER pathway genes cluster consistently in a specific region of the ranked list, accumulating individual small signals into a significant pathway-level enrichment score. Goal 2: Confirming stem cell population identity. GSEA tested whether entire published stem cell gene sets (BenPorath, Pece) were enriched in sorted versus parental cells, providing cross-study validation of population identity. Goal 3: Identifying racial differences. Comparison of NES values across 12 comparisons in the NES heatmap identified which pathways differ racially, in which direction, and under which conditions.

Partek Flow built-in GSEA vs. GSEAPreranked. Partek Flow includes a built-in GSEA tool that operates on normalized count data (median ratio or equivalent) and implements the standard Broad Institute walking-statistic enrichment algorithm. This is a legitimate pathway analysis approach and avoids the filtered-gene-list limitation of Partek's separate Gene Set Enrichment function (which applies Fisher's exact test to a pre-filtered significant gene list and would discard the sub-threshold NER signal entirely). However, Partek's built-in GSEA imposes a critical structural constraint for this study: it requires a factor with exactly two levels and therefore cannot accommodate a multi-factor interaction model. Running it on this dataset would require collapsing the four-variable design (ethnicity × treatment × cell type) into a series of isolated two-group comparisons, losing the ability to formally test whether arsenic acts differently across racial backgrounds or cell types simultaneously — which is precisely the interaction the study was designed to detect. This was confirmed

empirically: when Partek's built-in GSEA was run on DESeq2-normalized counts across three collapsed comparisons (AA vs. NHW, Arsenic vs. Vehicle, Stem vs. Parental), KEGG_NUCLEOTIDE_EXCISION_REPAIR showed no significant enrichment in any comparison (best result: NES = −1.31, FDR = 0.38 in the Stem vs. Parental comparison), compared to NES = −2.75, FDR = 0.18 in the C8-specific contrast from GSEAPreranked. When run on CPM-normalized counts instead, the signal degraded further (Stem vs. Parental NES = −0.56, FDR = 1.00). The primary finding is entirely absent from both Partek GSEA runs because collapsing all racial backgrounds and arsenic conditions into a single Stem vs. Parental comparison averages the AA-specific, arsenic-dependent NER suppression signal into noise. GSEAPreranked fed by DESeq2's full factorial model was therefore required: DESeq2 fits the complete interaction structure and produces per-gene statistics for each of the 12 pairwise contrasts, and GSEAPreranked consumes those pre-ranked lists, preserving the full model complexity at the pathway level. Partek's Gene Set Enrichment function (Fisher's exact test on a filtered gene list) was also tested; applied to the unfiltered all-gene dataset it returned a rich factor of 1.0 for NER (all 46 pathway genes present in the input list), rendering the result uninformative — it merely confirmed that NER genes exist in the dataset without any inference about differential expression or direction.[17]

Gene Set ANOVA vs. GSEA. Gene Set ANOVA[19] Partek Flow's Gene Set ANOVA uses a generalized linear model that fits expression across all genes in a pathway simultaneously, with Gene ID included as a model term to account for gene-to-gene baseline differences, and a Factor × Gene ID interaction term added automatically to test whether genes within the set show differential patterning across factor levels — what Partek terms a "disruption effect." A small p-value on the interaction term indicates that the expression pattern across pathway genes is disrupted by the factor, not merely shifted uniformly. While this is a more principled model than a simple F-test on average pathway expression, it still requires within-group replicates for variance estimation and therefore cannot be applied to any comparison involving the n=1 stem cell conditions (C5-C12). Gene Set ANOVA also weights all pathway genes equally, diluting signal from the small coherent core of regulated genes, whereas GSEA's weighted enrichment statistic weights each gene by its rank value, emphasizing the most significantly regulated members. Gene Set ANOVA is most appropriate for detecting disruption of expression patterning across pathway genes in well-powered balanced designs; in this study it was reserved as a potential confirmatory approach for the parental comparisons (C1-C4) where all groups have n=3, but was not the primary method for any comparison. Empirically, Gene Set ANOVA on DESeq2-normalized counts returned a non-significant main effect for NER in both the AA vs. NHW ($p = 0.80$) and Arsenic vs. Vehicle ($p = 0.49$) comparisons, and a modestly significant result in Stem vs. Parental ($p = 0.011$, FDR = 0.034), directionally consistent with but weaker than the GSEA finding. Notably, the disruption p-value for NER — testing whether gene patterning within the pathway differs — was extraordinarily significant in every comparison (Stem vs. Parental: $p < 10^{-18}$; AA vs. NHW: $p < 10^{-15}$), confirming genuine NER disruption across conditions but without the directional or condition-specific resolution needed to identify the AA + arsenic + stem cell combination as the driver. When Gene Set ANOVA was applied to CPM-normalized counts instead of DESeq2-normalized counts, NER appeared spuriously significant in the AA vs. NHW comparison ($p = 0.049$) despite a $\log_2$ fold change near zero (−0.003), illustrating the false positive inflation from composition bias. Across all three comparisons, CPM-based Gene Set ANOVA called 496 significant pathways ($p < 0.05$) versus 344 for DESeq2-based Gene Set ANOVA — a 44% inflation consistent with systematic composition bias generating signals without biological directionality.

Database selection rationale. The four databases were chosen because each provides a distinct and complementary function. Hallmark (50 gene sets) serves as a quality filter: its sets were computationally derived from thousands of overlapping literature gene sets, collapsed into 50 well-validated biological programs with minimal internal redundancy and cross-study validation. Signals appearing in Hallmark are robust by construction, but the database has no NER gene set because NER is too mechanistically specific for broad program-level curation. KEGG legacy organizes genes into mechanistic biochemical pathway maps with separate gene sets for NER, BER, MMR, and HR, making it the appropriate database to detect a specific NER suppression signal rather than a general DNA repair suppression — it draws boundaries around repair pathways at the correct biochemical resolution, which is why KEGG_NUCLEOTIDE_EXCISION_REPAIR was the

primary finding database. Reactome provides independent confirmation and mechanistic granularity: its hierarchical, independently curated annotation separates NER into sub-pathway steps (damage recognition, incision complex formation, dual incision, gap-filling synthesis, ligation), and agreement between KEGG and Reactome NER findings — built by different groups using different curation methods — constitutes strong evidence against a gene-set-definition artifact. Reactome also identified the COP9 signalosome suppression extending the impairment upstream of damage recognition, a finding absent from KEGG's schema. CGP (Chemical and Genetic Perturbations) is an empirical database of transcriptional signatures derived from published perturbation experiments. KAN_RESPONSE_TO_ARSENIC_TRIOXIDE, a gene set from an actual arsenic exposure experiment, resides in CGP and enabled cross-study validation that arsenic-exposed cells in the present study were activating programs consistent with published arsenic responses. CGP also contains the BenPorath and Pece stem cell signatures used to validate sorted population identity. Using all four databases simultaneously meant the 48 total GSEA runs could distinguish findings replicated across independently curated sources from those appearing in only one database — the convergence of KEGG and Reactome NER suppression in C8 and their simultaneous absence in C7 is the core evidentiary basis for the primary finding.

11. GSEA Implementation and Statistical Metrics

GSEAPreranked was used for all analyses. Settings: No_Collapse (HGNC symbols match MSigDB format), 1000 permutations, weighted enrichment statistic, minimum gene set size 15, maximum 500, fixed seed 149. Total: 48 runs (12 comparisons × 4 databases). Databases: Hallmark (50 curated gene sets, minimal overlap[23]); KEGG legacy (mechanistic pathway maps with separate NER, BER, MMR, HR gene sets[24]); Reactome (hierarchical pathway descriptions enabling identification of specific affected steps[25]); CGP (empirically derived signatures including KAN_RESPONSE_TO_ARSENIC_TRIOXIDE and BenPorath/Pece stem cell signatures for population validation.[20,21])

12. Ranked Gene Lists and the Rank Metric

For each of the 12 comparisons, a ranked gene list was constructed as: Rank Metric = $-\log_{10}$(p-value) × sign(shrunken log2 fold change) The $-\log_{10}$(p-value) component places the most statistically significant genes at the list extremes. The sign(shrunken L2FC) component encodes direction. Using the apeglm-shrunken fold change prevents lowly expressed genes with unreliable large fold changes from dominating extreme rank positions. Genes with p-values of zero were assigned ±300. This metric integrates both statistical significance and directionality from the DESeq2 model and is the established approach for GSEAPreranked with pre-computed statistics from an external factorial model.[26] Lists were exported as two-column tab-delimited .rnk files with Unix line endings.

Pipeline order and the DESeq2–GSEA relationship. DESeq2 ran first and completely before GSEA received any input; the two tools operate sequentially, not simultaneously or iteratively. DESeq2 took the raw count matrix (16,682 genes × 16 samples) and fit the full factorial model, producing for each of the 12 pairwise contrasts a results table containing a Wald test statistic, raw p-value, FDR-adjusted p-value (Benjamini-Hochberg), and apeglm-shrunken $\log_2$ fold change for all 16,682 genes. That results table was the direct input to GSEAPreranked. GSEAPreranked — rather than standard GSEA — was required because the factorial DESeq2 model is too complex for GSEA to fit internally from raw counts and a phenotype label; pre-computed per-gene statistics from the external model had to be supplied. Settings were: No_Collapse (HGNC symbols already matched MSigDB format, so no additional ID mapping was needed), 1,000 permutations, weighted enrichment statistic, minimum gene set size 15, maximum 500, fixed random seed 149. The minimum size of 15 excludes pathways too small for reliable statistical evaluation; the maximum of 500 prevents large, loosely defined gene sets from dominating results.

How enrichment analysis works. GSEA addresses a fundamental limitation of single-gene analysis for coordinated pathway changes: NER genes responding to arsenic show subtle shifts (1.2- to 1.8-fold) that individually never reach the FDR < 0.05 threshold after correcting for 16,682 tests, yet collectively reflect a coordinated suppression of the entire pathway. Rather than asking whether each NER gene is individually

significant, GSEA asks whether NER genes as a group behave consistently and non-randomly across the ranked gene list. The ranked list places the most significantly upregulated genes at the top (large positive rank metric) and the most significantly downregulated genes at the bottom (large negative rank). GSEA then walks down this list, incrementing a running sum each time a pathway gene is encountered and decrementing it for non-pathway genes; the peak deviation of this running sum from zero is the raw Enrichment Score (ES). If pathway members are concentrated at one end of the list, the score reaches a large positive or negative peak; if they are scattered randomly, increments and decrements cancel and the score hovers near zero. This is the key insight: a collection of genes each with individual p-values of 0.07–0.12 — all sub-threshold — can produce a strong enrichment signal if they are all consistently displaced in the same direction, because their combined positional clustering is non-random. The NES normalizes the ES by the mean from 1,000 permutations of that gene set, correcting for gene set size so NES values are comparable across pathways of different sizes and across all 12 comparisons. A NES of −2.75 means the observed score is 2.75 standard deviations more extreme than the permuted null, regardless of whether the gene set contained 20 or 200 members. The sign of the NES indicates direction relative to the numerator group: negative NES in C8 (AA Stem vs. Parental Arsenic) means NER genes are enriched at the bottom of the list — predominantly lower in AA stem cells than AA parental cells under arsenic. The leading edge subset, comprising the specific genes encountered before the enrichment score peak, identifies the pathway members most responsible for the signal; ERCC1 appearing in both the C8 and C12 leading edges means it is consistently among the most suppressed NER genes across two independent comparisons, making it the highest-confidence functional target.

No intermediate filtering between DESeq2 and GSEA. A critical methodological choice was to retain all 16,682 genes in every ranked list with no filtering based on differential expression results. The most common error in GSEA analysis is pre-filtering to only significant DEGs (e.g., FDR < 0.05 or |L2FC| > 0.58) before running GSEA, which defeats the purpose of the method: GSEA was invented specifically to detect coordinated signals in genes that individually do not reach significance thresholds, and the NER finding — where individual NER genes had fold changes of −0.45 to −0.67 that variably met or missed FDR < 0.05 — depends entirely on retaining all sub-threshold genes in the ranked list. Moreover, the rank metric itself already encodes significance: a non-significant gene with p-value 0.4 naturally ranks near the middle of the list where it contributes almost nothing to any enrichment score. Explicit removal is therefore redundant and discards information GSEA was already handling correctly. The only pre-GSEA filtering was the initial count filter applied during DESeq2 setup (≥10 summed raw counts across all 16 samples), which removes genes too lowly expressed for reliable dispersion estimation and is standard practice entirely independent of differential expression outcomes. The 1,806 genes with "—" placeholder symbols were also retained using their Ensembl IDs; these contribute to rank positions but do not score as members of any named pathway, effectively acting as uninformative background spacers — the correct behavior.

13. Significance Criteria and Statistical Metrics

Three GSEA metrics are reported. The Enrichment Score (ES) and Normalized Enrichment Score (NES): GSEA first calculates a raw ES by walking down the ranked gene list from most upregulated to most downregulated, tracking a running sum that increments when a pathway gene is encountered and decrements for non-pathway genes; the peak deviation of this running sum from zero is the ES. The NES is calculated by dividing the observed ES by the mean ES from 1000 permutations, normalizing for gene set size so NES values are comparable across pathways and across all 12 comparisons. Positive NES: pathway genes enriched at top of ranked list (higher in numerator group). Negative NES: enriched at bottom (higher in denominator). Minimum enrichment criterion: |NES| > 1.5. The Nominal p-value is the proportion of permuted enrichment scores exceeding the observed score for a single pathway; useful for ranking individual pathway assessments but not the primary significance criterion. The FDR q-value is the primary significance metric: the expected proportion of false positives among all gene sets achieving a given NES or better, calculated against the permutation-derived null distribution.[17] The standard GSEA threshold of FDR < 0.25 was applied, justified because: (1) it was validated in the original GSEA publication; (2) gene sets are not independent, reducing the

effective number of independent tests; and (3) the permutation-based scoring is itself conservative.[17] Primary significant findings required |NES| > 1.5 AND FDR < 0.25.

The leading edge subset comprises the genes before the enrichment score peak - those most responsible for the enrichment signal. Three statistics characterize it: tags% (percentage of pathway genes in the leading edge), list% (position of the enrichment peak as a proportion of the ranked list), and signal% (an integrated signal strength calculated as tags% × (1 − list%)). Leading edge analysis was performed through the GSEA Leading Edge module to identify genes shared between or unique to leading edges from different comparisons.

14. Gene Annotation Recovery

Of 16,682 genes, 1,806 carried the placeholder "—" in Partek Flow output due to absent HGNC symbols in Ensembl release 105. An INDEX MATCH lookup against a BioMart27 annotation table recovered symbols for a subset; unresolved genes retained their Ensembl identifier. All 16,682 genes were retained in the ranked list regardless of annotation status.

15. Data Availability and Reproducibility

All GSEA analyses used a fixed random seed (149) to ensure reproducibility. Raw FASTQ files and processed count matrices will be deposited in NCBI GEO upon manuscript acceptance. Complete DESeq2 results (all 12 pairwise contrasts, 16,682 genes) and GSEA pathway tables (all 48 runs) will be provided as supplementary data.

RESULTS

A cross-contrast overview of differential-expression magnitude and top pathway-level enrichment is provided in Tables 2 and 3; these summarize the 12 comparisons at a glance before the section-by-section narrative below.

Table 2. Differential-expression summary per contrast (C1–C12). Top up-regulated and down-regulated annotated protein-coding genes ranked by $\log_2$ fold change among genes passing FDR < 0.05. Gene count includes all genes passing FDR < 0.05 (including unannotated features). Full DESeq2 tables are provided as Supplementary Table S2.

Contrast	Description	n DEG (FDR < 0.05)	Top ↑ Gene (L2FC)	Top ↓ Gene (L2FC)
C1	AA vs. NHW — Parental, Vehicle	4,218	CD36 (+7.61)	CBSL (-7.40)
C2	AA vs. NHW — Parental, Arsenic	3,820	MTCO1P12 (+7.39)	CBSL (-7.59)
C3	NHW Arsenic vs. Vehicle — Parental	3,758	HMOX1 (+5.27)	RGS5 (-4.86)
C4	AA Arsenic vs. Vehicle — Parental	3,693	LINC01139 (+5.86)	MEST (-7.70)
C5	NHW Stem vs. Parental — Vehicle	3,332	FOSB (+6.23)	ELN (-5.42)
C6	AA Stem vs. Parental — Vehicle	3,349	PDPN (+5.46)	EPB41 (-6.93)
C7	NHW Stem vs. Parental — Arsenic	3,373	NPIPA7 (+3.54)	RNVU1-18 (-7.22)
C8	AA Stem vs. Parental — Arsenic	3,200	FNDC1 (+4.41)	PTX3 (-3.82)
C9	NHW Stem Arsenic vs. Vehicle	391	HMOX1 (+5.32)	DEPTOR (-4.58)
C10	AA Stem Arsenic vs. Vehicle	315	RNVU1-18 (+4.49)	DPT (-2.71)
C11	AA vs. NHW — Stem, Vehicle	2,403	CD36 (+9.03)	EPB41 (-7.25)
C12	AA vs. NHW — Stem, Arsenic	1,938	CD36 (+9.02)	LRRC17 (-8.57)

L2FC = $\log_2$ fold change; FDR = Benjamini–Hochberg adjusted p-value from DESeq2. "Top ↑ Gene" and "Top ↓ Gene" are the single annotated gene with largest positive and largest negative L2FC (among genes with FDR

< 0.05); positive L2FC indicates higher expression in the first group of the contrast (e.g., higher in AA for C1/C2/C11/C12; higher in Arsenic for C3/C4/C9/C10; higher in Stem for C5/C6/C7/C8). Unannotated probe IDs and LOC identifiers are excluded; the total DEG count includes all significantly differentially expressed features.

Table 3. GSEA top enriched pathway per contrast (C1–C12) across curated databases (KEGG, Reactome, Hallmark). The single top-ranked pathway by |NES| with FDR < 0.25 is reported; where no pathway reached FDR < 0.25, the pathway with largest |NES| is shown. CGP chemical/genetic-perturbation signatures are excluded from this summary but are retained in Supplementary Table S1. Positive NES = enriched in the first group of the contrast (e.g., higher in AA, Arsenic, or Stem); negative NES = enriched in the second group. The paper-referenced KEGG_NUCLEOTIDE_EXCISION_REPAIR finding in C8 (NES = −2.75, FDR = 0.18) and other pathway hits discussed in the text are tabulated in full in Supplementary Table S1.

Contrast	Description	Database	Top Enriched Pathway	NES	FDR
C1	AA vs. NHW — Parental, Vehicle	Hallmark	INFLAMMATORY RESPONSE	-1.69	0.183
C2	AA vs. NHW — Parental, Arsenic	Hallmark	ANDROGEN RESPONSE	-1.77	0.052
C3	NHW Arsenic vs. Vehicle — Parental	Reactome	RIBOSOME QUALITY CONTROL RQC COMPLEX EXTRACTS A...	+2.24	0.042
C4	AA Arsenic vs. Vehicle — Parental	Hallmark	SPERMATOGENESIS	-2.15	< 0.001
C5	NHW Stem vs. Parental — Vehicle	Reactome	RNA POLYMERASE III TRANSCRIPTION INITIATION FRO...	-2.71	< 0.001
C6	AA Stem vs. Parental — Vehicle	Reactome	CO INHIBITION BY PD 1	-2.58	< 0.001
C7	NHW Stem vs. Parental — Arsenic	Hallmark	OXIDATIVE PHOSPHORYLATION	+2.76	< 0.001
C8	AA Stem vs. Parental — Arsenic	Reactome	MET PROMOTES CELL MOTILITY	-2.82	< 0.001
C9	NHW Stem Arsenic vs. Vehicle	Reactome	RIBOSOME QUALITY CONTROL RQC COMPLEX EXTRACTS A...	+2.60	< 0.001

Contrast	Description	Database	Top Enriched Pathway	NES	FDR
C10	AA Stem Arsenic vs. Vehicle	Hallmark	OXIDATIVE PHOSPHORYLATION	+2.67	< 0.001
C11	AA vs. NHW — Stem, Vehicle	Hallmark	INFLAMMATORY RESPONSE	-2.30	< 0.001
C12	AA vs. NHW — Stem, Arsenic	Reactome	SIGNALING BY FGFR2 IIIA TM	-2.23	< 0.001

NES = normalized enrichment score; FDR = false discovery rate (q-value). Databases: KEGG (canonical signaling/metabolic pathways); Reactome (biological processes and events); Hallmark (MSigDB distilled hallmark signatures). CGP = Chemical & Genetic Perturbation (MSigDB c2.cgp).

Section 1. Racial transcriptional differences in parental breast tissue-derived cells and their modulation by arsenic (C1–C4)

Widespread transcriptional differences distinguish BRL-23 and BRL-24 at baseline

Differential expression analysis of vehicle-treated parental cells (C1: AA vs. NHW Parental Vehicle) identified 7,355 significantly differentially expressed genes (FDR < 0.05) from 16,682 expressed genes, representing 44% of the transcriptome. Expression was higher in BRL-24 (AA) for 3,847 genes and in BRL-23 (NHW) for 3,508 genes. GSEA identified 71 significantly enriched pathways (|NES| > 1.5, FDR < 0.25). The AA line showed higher oxidative phosphorylation (HALLMARK_OXIDATIVE_PHOSPHORYLATION: NES = +1.68, FDR = 0.18); NHW cells showed higher inflammatory signaling (HALLMARK_INFLAMMATORY_RESPONSE: NES = −1.69, FDR < 0.25; HALLMARK_TNFA_SIGNALING_VIA_NFKB: NES = −1.57). BRL-24 expressed higher ALDH1A3, ITGA6, XPC, and ERCC2; BRL-23 expressed higher SNAI2, VIM, CD44, MSH2, PARP1, and DDB1. Although individual NER genes were higher in AA cells, KEGG_NUCLEOTIDE_EXCISION_REPAIR showed no significant enrichment (NES = +0.98, FDR = 0.72), indicating these individual gene differences do not reflect coordinated NER pathway activity at baseline. The KAN_RESPONSE_TO_ARSENIC_TRIOXIDE gene set trended toward higher expression in NHW cells at baseline (C1: NES = −1.54), suggesting BRL-23 cells are constitutively more transcriptionally primed for arsenic response (Figure 1, Figure 2A, Figure 3A,B).

Arsenic dramatically narrows racial transcriptional differences in parental cells

Under arsenic conditions (C2: AA vs. NHW Parental Arsenic), a 68% collapse in differentially expressed genes was observed, falling from 7,355 to 2,328. More notably, all 57 significantly enriched pathways in C2 were enriched in the NHW direction. Key genes reversed direction under arsenic: ALDH1A3, significantly higher in AA at baseline, shifted toward NHW under arsenic, as did SNAI2 and TGFBR1. Pathways persisting as NHW-enriched included HALLMARK_INFLAMMATORY_RESPONSE (NES = −1.77), HALLMARK_TNFA_SIGNALING_VIA_NFKB (NES = −1.76), and HALLMARK_XENOBIOTIC_METABOLISM (NES = −1.59) (Figure 1, Figure 2B).

BRL-23 and BRL-24 mount divergent transcriptional responses to arsenic

Within-line arsenic responses (C3: NHW Arsenic vs. Vehicle; C4: AA Arsenic vs. Vehicle) showed BRL-23 mounting a substantially larger response: 705 significant pathways in NHW versus 334 in AA (2.1-fold difference). NHW cells strongly induced inflammatory and stress programs (HALLMARK_INFLAMMATORY_RESPONSE: NES = +2.14; HALLMARK_HYPOXIA: NES = +2.05) and suppressed EMT (HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION: NES = −1.70). In AA cells, arsenic induced XPC (L2FC = +0.78, FDR < 0.001) with no equivalent in NHW (L2FC = +0.03, FDR = 0.87), suggesting divergent DNA damage response strategies. Both lines showed ERCC6 downregulation (BRL-23: L2FC = −0.29, FDR = 0.01; BRL-24: L2FC = −0.53, FDR < 0.001) and Hedgehog induction (C3: NES = +2.00; C4: NES = +1.54). The KAN_RESPONSE_TO_ARSENIC_TRIOXIDE gene set was significantly enriched in AA under arsenic (C4: NES = −1.63, FDR < 0.25) with leading edge genes including HMOX1, AKR1C1, SLC7A11, and CXCL8 (Figure 1, Figure 2C,D, Figure 3C).

Section 2. Transcriptional validation of stem cell-enriched populations (C5-C8)

To enrich for CSC-like populations, BRL-23 and BRL-24 cells were sorted using the CD24−/CD49f+/CD44+ surface phenotype. Comparisons of sorted versus parental cells under vehicle conditions (C5: NHW Stem vs. Parental Vehicle; C6: AA Stem vs. Parental Vehicle) confirmed population identity through three independent lines of evidence. First, principal component analysis showed PC1 (81% of total variance) completely separating stem from parental samples across both lines and conditions, confirming cell type as the dominant source of transcriptional variation. Second, GSEA identified 900 significant pathways in NHW (C5) and 877 in AA (C6) stem versus parental comparisons, far exceeding racial differences under any condition. Third, sorted cells showed coordinate enrichment of canonical stemness programs: HALLMARK_G2M_CHECKPOINT (C5: NES = +2.76), HALLMARK_E2F_TARGETS (NES = +2.75), BENPORATH_ES_2 (NES = +1.84), and PECE_MAMMARY_STEM_CELL_UP (NES = +1.85). Enrichment of the BenPorath and Pece signatures, derived from independently sorted stem cell populations in published studies, provides cross-study validation that the CD24−/CD49f+/CD44+ fraction represents a genuine stem cell transcriptional state (Figure 1, Figure 2E,F).

DNA repair capacity was broadly elevated in sorted stem cells, with significant enrichment of REACTOME_BASE_EXCISION_REPAIR, REACTOME_MISMATCH_REPAIR, and KEGG_HOMOLOGOUS_RECOMBINATION across C5-C8, providing functional validation of the sorted populations independent of surface markers. Individual repair genes LIG1 (C7: L2FC = +0.99, FDR < 0.001) and POLD1 (C7: L2FC = +1.42, FDR < 0.001) were elevated in NHW stem cells; CDK7 was consistently downregulated (C5-C8: L2FC −0.39 to −0.61, FDR < 0.05 in C5 and C8) (Figure 1, Figure 3A).

NER is specifically suppressed in AA stem cells under arsenic — the primary racial disparity finding

Comparisons of stem versus parental cells under arsenic (C7: NHW Stem vs. Parental Arsenic; C8: AA Stem vs. Parental Arsenic) revealed a pronounced racial divergence in NER. In AA stem cells, KEGG_NUCLEOTIDE_EXCISION_REPAIR was significantly suppressed (C8: NES = −2.75, FDR = 0.18), the single strongest pathway finding in the entire dataset, confirmed by Reactome (REACTOME_NUCLEOTIDE_EXCISION_REPAIR: NES = −2.68, FDR < 0.25). No equivalent suppression was observed in NHW (C7: NES = +0.86, FDR = 0.64), nor in any parental comparison (C1-C4 NES range +0.80 to +1.10, all FDR > 0.25), establishing that the racial NER disparity requires both stem cell identity and arsenic exposure.

Leading edge analysis of C8 identified 16 suppressed genes in two functional groups: damage recognition and TFIIH loading factors (XPC L2FC = −0.45, FDR = 0.016; DDB1 L2FC = −0.33, FDR = 0.035; CETN2 L2FC = −0.47, FDR = 0.014; RAD23A L2FC = −0.46, FDR = 0.016; ERCC3; GTF2H1; CDK7 L2FC = −0.61, FDR = 0.005) and gap-filling and ligation machinery (POLD2, POLE3, RFC1, RFC2, RPA2, RPA3). The Reactome C8 leading edge additionally included eight COP9 signalosome subunits (COPS3-8), extending suppression to chromatin remodeling steps preceding damage recognition. Individual gene-level analysis of C8 confirmed that

13 of 23 NER genes individually reached FDR < 0.05 in the DESeq2 per-gene results, all suppressed (CDK7 FDR = 0.00027, RFC1 FDR = 0.0015, ERCC1 FDR = 0.0089, XPC FDR = 0.016, RAD23A FDR = 0.019, CETN2 FDR = 0.031, POLD2 FDR = 0.048, among others), providing gene-level corroboration of the pathway-level signal. The contrast with C7 at the individual gene level is biologically notable: in NHW stem cells under arsenic, the gap-filling and ligation machinery goes in the opposite direction, with POLD1 (FDR = 2.3×10^{-7}, shrunken L2FC = +1.42), LIG1 (FDR = 0.00017, +0.99), RFC3 (FDR = 0.00043, +1.14), and POLE (FDR = 0.00011, +0.87) all significantly induced in NHW stem cells — the same genes suppressed in AA stem cells. ERCC1 is the only NER gene suppressed in both lines under arsenic in the stem cell comparisons (C7 shrunken L2FC = −0.54; C8 = −0.47), reinforcing its designation as the primary candidate for functional validation. NHW stem cells under arsenic (C7) showed the opposite pathway-level pattern, with REACTOME_BASE_EXCISION_REPAIR significantly elevated (NES = −2.43), indicating opposite DNA repair trajectories: NHW stem cells maintain BER while AA stem cells suppress NER (Figure 1, Figure 4A).

Two-database convergence strengthens the biological credibility of this finding. KEGG and Reactome are curated independently by different organizations using different annotation philosophies (KEGG organizes genes by biochemical pathway map; Reactome organizes genes by reaction-hierarchy with explicit sub-pathway steps), and they define the NER gene set from non-overlapping source literature. Both NER gene sets returned highly significant negative enrichment in C8 (KEGG NES = −2.75, FDR = 0.18; Reactome NES = −2.68, FDR < 0.25), while neither returned a significant NER hit in C7 (KEGG NES = +0.86, FDR = 0.64; Reactome NES = +0.74, FDR = 0.58) or in any of the four parental-cell comparisons (C1–C4; all |NES| < 1.10 and all FDR > 0.25). Convergence of two independently curated NER gene sets in the same comparison, together with the non-overlap of Reactome's COP9 signalosome module with KEGG's core NER set, argues against a gene-set-definition artifact and supports the conclusion that the C8 NER suppression reflects a genuine coordinated transcriptional program rather than a single-database idiosyncrasy.

Section 3. Arsenic-induced transcriptional responses within stem cell populations (C9-C10)

Within-stem arsenic comparisons (C9: NHW Stem Arsenic vs. Vehicle; C10: AA Stem Arsenic vs. Vehicle; both n=1, pathway-level GSEA reported) identified 858 significant pathways in C9 and 967 in C10 - the highest pathway count in the dataset. Arsenic induced a shared program in both lines: HALLMARK_OXIDATIVE_PHOSPHORYLATION (C9: NES = +2.51; C10: NES = +2.67), HALLMARK_HEDGEHOG_SIGNALING (C9: NES = +2.15; C10: NES = +2.01), HALLMARK_NOTCH_SIGNALING (C9: NES = +1.79; C10: NES = +1.70), HALLMARK_HYPOXIA (C9: NES = +2.19; C10: NES = +2.35), and HALLMARK_MYC_TARGETS_V1 (C9: NES = +2.05; C10: NES = +1.66). The NFE2L2 antioxidant axis was activated in both lines with HMOX1 and SLC7A11 among the highest-ranked genes. Critically, KEGG_NUCLEOTIDE_EXCISION_REPAIR was not significantly altered within either stem line under arsenic (C9: NES = +1.08, FDR = 0.99; C10: NES = +0.85, FDR = 0.33), establishing that the NER suppression in C8 and C12 reflects a cell-intrinsic NER expression difference between sorted AA stem cells and their parental counterparts rather than arsenic directly suppressing NER. Senescence and apoptosis programs were not significantly activated in sorted stem cells under arsenic from either line (Figure 1, Figure 2I,J).

Section 4. Racial transcriptional differences are amplified in the stem cell compartment (C11-C12)

Stem cell enrichment amplifies rather than homogenizes racial differences

Direct racial comparisons within sorted stem cells (C11: AA vs. NHW Stem Vehicle; C12: AA vs. NHW Stem Arsenic) revealed dramatic amplification of racial transcriptional differences. C11 identified 582 significantly enriched pathways and C12 identified 503 significant pathways, compared to only 71 (C1) and 57 (C2) in parental racial comparisons - a roughly tenfold amplification consistent across all four MSigDB databases under both conditions (Figure 1, Figure 2K,L).

NER racial disparity confirmed and shown to require arsenic

In C12, KEGG_NUCLEOTIDE_EXCISION_REPAIR was significantly enriched in NHW cells (NES = −1.87, FDR < 0.25; Reactome: NES = −2.16), confirming lower NER in AA stem cells under arsenic. NER was not significantly different without arsenic (C11: NES = +0.63, FDR = 0.59) - it is the only major pathway absent in C11 and emerging in C12, establishing arsenic as the specific trigger that unmasks the racial NER disparity. The C12 leading edge identified 17 suppressed AA genes comprising gap-filling and ligation machinery: PCNA, LIG1 (L2FC = −0.74), POLD1, POLD3, POLE, POLE2, RFC3-5. Cross-comparison of C8 and C12 KEGG NER leading edges identified four genes present in both - CCNH, ERCC1, RFC2, RPA3 - the most reproducible markers of the racial NER disparity. ERCC1 showed consistent suppression in AA stem cells across C5-C8 (L2FC −0.44 to −0.67, FDR < 0.05), establishing it as the primary candidate for functional validation (Figure 1, Figure 4A).

NHW cells show higher inflammatory signaling, xenobiotic capacity, and arsenic response priming

The NHW inflammatory advantage was preserved and amplified in stem cells: HALLMARK_INFLAMMATORY_RESPONSE was significantly enriched in NHW stem cells in C11 (NES = −2.30, FDR < 0.05) and C12 (NES = −1.99, FDR < 0.05), substantially stronger than in parental comparisons (C1: NES = −1.69). Cross-comparison identified 23 core inflammatory genes shared across C1 and C11 including ICAM1, IL1A, IL1R1, IL7R, IFITM1, LYN, and TLR3. Eighteen additional genes were gained in C11 stem cells including IL6, CXCL8, CCL2, NFKBIA, and IRAK2. NHW stem cells also showed significantly higher xenobiotic metabolism (C11: NES = −1.82) and constitutively higher KAN_RESPONSE_TO_ARSENIC_TRIOXIDE expression (C11: NES = −1.69; C12: NES = −1.65).

AA cells maintain higher oxidative phosphorylation

The AA OxPhos advantage was the most stable racial difference in the dataset: HALLMARK_OXIDATIVE_PHOSPHORYLATION was significantly enriched in AA cells in C1 (NES = +1.68), C2 (NES = +1.26), C11 (NES = +1.77, FDR < 0.25), and C12 (NES = +1.72, FDR < 0.25), unaffected by arsenic treatment or stem cell enrichment (Figure 1, Figure 3C).

Integrative summary

Across all 12 comparisons, this study identifies a layered racial transcriptional landscape qualitatively transformed in the CSC-like compartment. In parental cells, arsenic dramatically narrows racial differences (68% collapse in DEGs), yet residual differences uniformly favor NHW cells. In stem cells, racial differences are amplified tenfold and prove more resistant to arsenic-driven convergence. Three stable themes predominate: higher OxPhos in AA cells, higher inflammatory and xenobiotic signaling in NHW cells, and a constitutively more arsenic-primed transcriptional state in NHW cells. Against this backdrop, a fourth theme emerges specifically at the intersection of stem cell identity and arsenic exposure: arsenic unmasks a selective suppression of NER in AA stem cells - affecting both initiation (XPC, DDB1, ERCC3, GTF2H1) and completion (PCNA, LIG1, RFC3-5) - absent under all other conditions. ERCC1 is the most reproducible single-gene marker of this disparity. These findings identify the CSC-like stem cell compartment as the primary locus of racial biological divergence and provide a transcriptomics-level mechanistic framework linking AA racial background, arsenic exposure, stem cell biology, and impaired genomic maintenance.

DISCUSSION

Breast cancer racial disparities represent one of the most persistent and deadly inequities in cancer outcomes: African American women experience approximately 40% higher breast cancer mortality than non-Hispanic White women, a gap that has widened over recent decades despite advances in treatment.[29,30] Environmental arsenic exposure is an established carcinogen with evidence for racial variation in both environmental levels and tissue-level sensitivity.[4,32] Despite these observations, the cellular and molecular mechanisms by which racial background modifies vulnerability to arsenic-induced genomic damage have remained entirely uncharacterized. The present study addresses this gap through genome-wide transcriptional profiling of two patient-derived breast tissue cell lines across 12 pairwise comparisons spanning parental and CD24−/CD49f+/CD44+ sorted CSC-like populations, vehicle and arsenic conditions, and within-line and cross-line contrasts. The central finding - that arsenic unmasks a selective suppression of NER in African American CSC-like cells absent in non-Hispanic White stem cells under identical conditions - provides the first transcriptomics-level mechanistic framework connecting race, arsenic exposure, breast tissue-derived CSC-like biology, and impaired genomic maintenance in a cancer-relevant model system.

Arsenic as a co-carcinogen that inhibits NER: placing the current findings in mechanistic context

Arsenic is classified as a Group 1 human carcinogen, yet its carcinogenic mechanisms differ fundamentally from classical mutagens.[32] Rather than causing characteristic point mutations, arsenic acts primarily as a co-carcinogen by amplifying the genotoxic effects of other environmental insults.[33,34] A key mechanism is its capacity to impair NER, the primary pathway removing helix-distorting lesions induced by UV radiation, polycyclic aromatic hydrocarbons, and arsenic-associated oxidative adducts.[33,35] Arsenic inhibits NER through multiple mechanisms: it reduces XPC and DDB2 expression,[35,36] displaces zinc from the zinc finger domains of NER proteins XPA and PARP-1,[37,38] and suppresses p53-mediated transcriptional activation of downstream NER genes.[39] The racial specificity of the NER suppression identified here - specific to the AA CSC-like context, not observed in NHW stem cells under identical conditions, not observed in parental cells of either racial background under any condition - was not anticipated by prior work, which studied arsenic-NER interactions in race-indeterminate bulk cell populations. The suppression operates at both initiation and completion stages simultaneously, with COP9 signalosome suppression[40] extending the failure to chromatin remodeling upstream of damage recognition.

ERCC1 as the keystone marker of racial NER disparity

ERCC1, encoding the obligate partner of XPF in the 5′ NER incision endonuclease complex, is functionally indispensable for both GG-NER and TC-NER.[41,42] ERCC1 is catalytically inactive alone, functioning by positioning the ERCC1-XPF complex at the pre-incision bubble through interactions with XPA and RPA.[41,43] Its consistent suppression across all four AA stem cell comparisons (C5-C8: L2FC −0.44 to −0.67, FDR < 0.05) and presence in both the C8 and C12 KEGG NER leading edges makes it the single most reproducible gene-level marker of the racial NER disparity. Clinically, high ERCC1 expression is associated with platinum chemotherapy resistance.[41] The present findings suggest a complementary implication: low ERCC1 in AA CSC-like cells under arsenic may predict elevated susceptibility to arsenic-induced DNA damage accumulation.

Cancer stem cells as the primary locus of racial biological divergence

The tenfold amplification of racial pathway differences in stem cells relative to parental cells (C11: 582 vs. C1: 71; C12: 503 vs. C2: 57) is the most architecturally striking finding of this study. CSC-like cells defined by the CD24−/CD49f+/CD44+ phenotype represent a well-validated progenitor-enriched population with self-renewal capacity, elevated DNA repair programs, and resistance to genotoxic stress.[6,7] The sorted fractions displayed canonical stem cell transcriptional profiles confirming their biological identity, including BenPorath and Pece breast tissue-derived CSC-like signatures,[20,21] elevated BER, MMR, and HR programs,[22] and Notch, Hedgehog, and hypoxia pathway activation. The arsenic dependence of the racial NER disparity - absent in C11, present in C12 - demonstrates that environmental exposure is required to unmask a latent vulnerability in AA stem cells, consistent with the gene-environment interaction model of cancer health disparities.[44]

Expected, unexpected, and novel findings

Several results confirmed predictions based on the existing literature and provided internal study validation. The AA OxPhos advantage (higher across all 12 comparisons) is consistent with published evidence of population-level differences in mitochondrial metabolic gene expression. Hedgehog induction by arsenic (C3, C4, C9, C10) is consistent with published evidence that arsenic promotes CSC-like expansion through GLI-dependent programs[45,46] while acting as a GLI antagonist in bulk cells. Stem cell elevation of BER, MMR, and HR relative to parental cells is textbook CSC-like biology[22] and served as internal validation that CD24−/CD49f+/CD44+ sorting successfully isolated a genomically protected, repair-competent population. Acute EMT suppression by arsenic in NHW cells is well documented in the short-term exposure literature. NHW constitutive arsenic response priming (KAN arsenic response genes higher at baseline in C1) explains the consistently larger NHW transcriptional response to arsenic.

Three findings were unexpected and represent novel observations. First, the 68% collapse in racial differences under arsenic in parental cells ran counter to the common assumption that a genotoxic stressor would amplify pre-existing racial biological differences. The shared arsenic stress response homogenized both transcriptomes, with all residual differences uniformly favoring NHW cells - arsenic acted as a transcriptional equalizer at the bulk population level while simultaneously creating new disparities in specific cell subpopulations. Second, the tenfold amplification of racial differences in stem cells was unexpected. Stem cells might intuitively be expected to be more similar across populations as an earlier, less differentiated cellular state. The opposite is true: the sorting procedure strips away shared bulk transcriptional programs, revealing a deeper layer of racial biology that is more pronounced in the stem compartment. This implies that bulk-population disparity studies may systematically underestimate the magnitude of biologically meaningful racial differences. Third, the strict three-condition requirement for the racial NER disparity - requiring African American background AND stem cell identity AND arsenic exposure simultaneously - was not anticipated by prior work, which did not distinguish racial backgrounds or examine stem cells. NER is the only major pathway to show the specific pattern of absence in C11 and emergence in C12, making this biologically unique in the dataset.

Arsenic maintains self-renewal programs in stem cells of both racial backgrounds

The consistent induction of Hedgehog and Notch by arsenic in sorted stem cells (C9, C10) appears paradoxical given that arsenic is known to inhibit GLI1 directly in bulk cancer cell lines.[45,46] In sorted stem cells, Hedgehog (NES = +2.15 in NHW, +2.01 in AA) and Notch (NES = +1.79 and +1.70) were significantly induced. This may reflect population dynamics: if arsenic selectively eliminates non-stem cells with high Hedgehog activity while sparing stem cells, the within-sorted-population comparison appears as induction. Non-canonical GLI activation[47] may also operate differently in stem cells. The absence of significant apoptosis or senescence in sorted stem cells under arsenic from either line[22] combined with maintained self-renewal creates conditions under which arsenic-exposed AA stem cells - carrying suppressed NER and elevated OxPhos-driven ROS - could accumulate mutations over time without cytotoxic elimination.

Racial differences in inflammatory signaling and their potential significance

The consistently higher NHW inflammatory tone (23-gene core signature amplified in stem cells) is counterintuitive given that chronic inflammation is a known breast cancer driver and is associated with worse outcomes in African American patients.[48] The NHW inflammatory signature - dominated by interferon-stimulated genes (IFITM1), pattern recognition receptors (TLR3), and lymphocyte activation markers (IL7R, LYN) - may represent immune surveillance capacity rather than a classically pro-tumorigenic cytokine state. The amplification in NHW stem cells, adding IL6, CCL2, CXCL8, NFKBIA, and IRAK2, is consistent with the known role of NFκB signaling in CSC-like self-renewal and maintenance.[49]

The arsenic convergence paradox and integrated mechanistic model

The 68% collapse of racial differences under arsenic, with all residual differences favoring NHW cells, reflects the dominance of the shared arsenic stress response over racial biology in the bulk population. The

KAN_RESPONSE_TO_ARSENIC_TRIOXIDE gene set, already enriched in NHW cells at baseline (C1: NES = −1.54), indicates constitutive NHW arsenic transcriptional priming, explaining both the larger NHW arsenic response (705 vs. 334 pathways) and the uniform residual NHW-direction differences. In stem cells, this convergence fails and the NER racial disparity emerges, reinforcing that arsenic-race interactions are fundamentally different in the stem cell compartment.

An integrated mechanistic model: At baseline, AA stem cells have lower NER gene expression relative to their parental counterparts (reflected across C5-C8). Upon arsenic exposure, NHW stem cells maintain or upregulate BER while AA stem cells' already-lower NER expression falls below the threshold for racial significance, producing the arsenic-dependent racial NER divergence. Arsenic-induced zinc finger protein displacement[37,38] and p53 transcriptional suppression[39] may have larger quantitative effects in cells already expressing less NER machinery. AA stem cells' constitutively higher OxPhos activity generates elevated mitochondrial ROS; arsenic generates additional ROS.[32,34] Higher oxidative DNA lesion burden alongside reduced NER capacity creates a mutagenic double-hit specific to AA stem cells under arsenic - a mechanism consistent with observed racial disparities in breast cancer outcomes and therapy resistance.

Limitations and future directions

The most significant limitation is n=1 per condition for sorted stem cell populations (C5-C12), necessitated by sorting yield constraints. GSEA is more robust to n=1 than individual gene statistics because pathway-level enrichment aggregates signals across 15-500 genes simultaneously, but individual gene-level conclusions from stem cell comparisons should be treated as hypothesis-generating. The NER pathway findings are robust because they represent coordinated suppression of an entire biological program (NES = −2.75, 16 KEGG genes all in the same direction). Future studies with triplicate stem cell replicates would enable a three-way interaction analysis (ethnicity × treatment × cell type) providing the most direct statistical test of the central hypothesis.

Second, this study uses two patient-derived breast tissue cell lines as proxies for NHW and AA racial biology. Validation in primary breast epithelial cells, organoids, and breast tumor specimens from racially characterized patients is required to determine whether the NER and inflammatory differences are characteristics of the broader racial backgrounds or are idiosyncratic to BRL-23 and BRL-24. Third, the study examines transcriptional abundance, not protein function or actual DNA repair capacity. Direct measurement of NER-specific lesion removal kinetics[35] in sorted populations under arsenic is an essential next step. Finally, why AA stem cells have lower baseline NER gene expression than NHW stem cells remains unresolved. Epigenetic differences at NER gene promoters, given arsenic's well-documented effects on DNA methylation[50] and the known regulation of XPC by CpG methylation, are priority candidates for investigation.

CONCLUSIONS

This study provides the first transcriptomics-level evidence that African American and non-Hispanic White patient-derived breast tissue cell lines differ fundamentally in their NER pathway response to arsenic within the CSC-like compartment. The analytical pipeline - DESeq2 with a full interaction model followed by GSEAPreranked across four MSigDB databases - was specifically designed to detect accumulated small changes in NER and DNA repair pathway gene expression and to validate stem cell population identity through enrichment of canonical signatures. Arsenic unmasks a selective and reproducible NER suppression in AA stem cells affecting both initiation (XPC, DDB1, CETN2, RAD23A, ERCC3, GTF2H1, CDK7, COP9 signalosome) and completion (PCNA, LIG1, RFC3-5, POLD1, POLE) stages, requiring the specific combination of African American cellular biology, CD24−/CD49f+/CD44+ stem cell identity, and arsenic exposure. ERCC1 is the primary single-gene candidate for functional validation. The tenfold amplification of racial differences in the stem compartment, with NER uniquely unmasked by arsenic, identifies the CSC-like stem cell as the primary locus of racial biological divergence and provides a mechanistic framework directly relevant to racial disparities in arsenic-associated breast cancer risk.

These findings motivate: functional validation of NER capacity in primary AA and NHW breast epithelial cells and organoids under arsenic exposure; ERCC1 protein quantification in breast tumor specimens stratified by racial background; generation of additional stem cell replicates to enable the three-way interaction statistical tests that would provide the most direct test of the central hypothesis; and prospective epidemiological studies examining NER gene expression as a biomarker of arsenic-associated breast cancer risk in diverse populations.

SUPPLEMENTARY MATERIALS

The following supplementary materials accompany this manuscript and are organized in the companion project folder Arsenic NER Project Deliverables/ (see 00_README.txt for full layout). File names listed in brackets below correspond to the renamed, reorganized deliverables folder.

Supplementary Table S1. Complete GSEA results for all 48 runs (12 comparisons C1–C12 × 4 MSigDB databases: Hallmark, KEGG legacy, Reactome, CGP). Contains NES, nominal p-value, FDR q-value, FWER p-value, enrichment score, leading-edge percentage (tags/list/signal), and gene-set size for every pathway returned. Organized as one tab per contrast with database as a filter column, plus a long-format summary tab for cross-contrast comparison. *[File: S1_GSEA_Results_All_48_Runs.xlsx]*

Supplementary Table S2. Full DESeq2 differential-expression tables for each of the 12 pairwise contrasts (C1–C12). Includes base mean, apeglm-shrunken $\log_2$ fold change, standard error, Wald statistic, nominal p-value, Benjamini–Hochberg adjusted FDR, and HGNC gene-symbol annotation for all 16,682 tested genes per contrast. Organized as 12 tabs (one per contrast) plus a long-format tab for cross-contrast queries. *[File: S2_DESeq2_Results_C1-C12.xlsx]*

Supplementary Table S3. Leading-edge gene inventories for the pathways called out in the main text — KEGG_NUCLEOTIDE_EXCISION_REPAIR, REACTOME_NUCLEOTIDE_EXCISION_REPAIR, HALLMARK_INFLAMMATORY_RESPONSE, HALLMARK_TNFA_SIGNALING_VIA_NFKB, HALLMARK_OXIDATIVE_PHOSPHORYLATION — across C1, C7, C8, C11, and C12. Each row lists gene symbol, rank metric, rank position, core-enrichment flag, and the group categorization used in Figure 4 (Shared / C8-only / C12-only / additional NER; Shared C1+C11 / C11-only / C1-only). *[File: S3_Leading_Edge_Gene_Lists.xlsx]*

Supplementary Figure S1. Full 42-pathway NES heatmap extending Figure 1 to include all four MSigDB databases (Hallmark, KEGG, Reactome, CGP) across the 12 comparisons, with a pathway-level source-database annotation column. *[File: Figure_S1_42-Pathway_NES_Heatmap.jpg]*

Supplementary Figure S2. Per-comparison principal component analysis (PCA) of DESeq2 rlog-transformed counts, with 95% confidence ellipses per treatment group, demonstrating the separation of parental-vs-stem and vehicle-vs-arsenic axes that anchor the factorial interaction model. This is the reference PCA corresponding to the schematic shown on the companion interactive site. *[File: Figure_S2_PCA_Plots.jpg]*

Supplementary Figure S3. Stem-cell-signature enrichment validation plots showing GSEA enrichment curves for BENPORATH_ES_1, PECE_MAMMARY_STEM_CELL_UP, and LIM_MAMMARY_STEM_CELL_UP signatures in sorted-vs-parental comparisons (C5, C6), confirming genuine CSC-like population identity of the CD24−/CD49f+/CD44+ sorted fraction. *[File: Figure_S3_Stem_Signature_Validation.jpg]*

Supplementary Figure S4. XPC and XPA expression comparison across the 12 contrasts, rendered as side-by-side bar strips with shrunken $\log_2$ fold change, FDR annotation, and cell-line/treatment grouping. Provides gene-level detail to complement the KEGG_NUCLEOTIDE_EXCISION_REPAIR pathway-level findings. *[File: Figure_S4_XPC_XPA_Comparison.jpg]*

Supplementary Figure S5. Side-by-side comparison of the four analytic pipelines evaluated for this study (Partek built-in GSEA, GSEAPreranked on DESeq2 shrunken L2FC, Partek Gene Set ANOVA, and CPM + $\log_2$FC ranking), with concordance annotations for key findings. *[File: Figure_S5_Analysis_Pipeline_Comparison.jpg]*

Supplementary Data S1. Raw HTSeq gene-level count matrix (all 16 samples × 60,675 Ensembl gene features), DESeq2 variance-stabilized and rlog-transformed normalized count matrices, and sample metadata (.tsv) with sample ID, cell line, donor race, treatment, population, replicate, and contrast assignments. Provided for full independent reanalysis. *[File: Data_S1_Count_Matrices/ (folder)]*

Supplementary Data S2. CPM-normalized count matrix (the pre-DESeq2 normalization output produced by Partek) provided for pipeline transparency and to allow reviewers to reproduce the CPM vs. DESeq2 comparison discussed in Methods Sections 8–9. *[File: Data_S1_Count_Matrices/CPM_Normalized_Counts.txt]*

Supplementary Data S3. Gene Set ANOVA results (Partek built-in) for the three main effects modeled in the factorial design (AA vs. NHW, Arsenic vs. Vehicle, Stem vs. Parental). Provided as three tab-delimited tables and discussed in Supplementary Methods alongside the GSEAPreranked results. *[File: Reproducibility/Alternative_Analyses/]*

Supplementary Data S4. Pre-ranked input files for GSEAPreranked (.rnk format, gene symbol and DESeq2 rank metric) for all 12 contrasts, enabling exact reproduction of the GSEA results in Supplementary Table S1. *[File: Reproducibility/Pre-ranked_Input_Files/]*

Supplementary Data S5. Full GSEAPreranked output directories for all 48 runs (12 contrasts × 4 databases), including per-pathway HTML reports, enrichment-plot SVGs, leading-edge gene indices, and the gene-level rank metric used for each run. *[File: Reproducibility/GSEA_Per-Contrast_Reports/]*

Supplementary Methods. Expanded comparison narrative for (i) CPM vs. DESeq2 normalization choice (including the rationale for preferring DESeq2's size-factor-based normalization and shrunken L2FC for GSEAPreranked ranking), (ii) Partek built-in GSEA vs. GSEAPreranked (with specific NES values for the C8 KEGG_NER comparison under both approaches), and (iii) Gene Set ANOVA vs. GSEA (discussing why Gene Set ANOVA is appropriate for main-effect hypotheses but under-powered for the interaction-style contrasts that drive this study's conclusions). Includes parameter settings referenced in Methods Sections 8–10. *[File: Supplementary_Methods.docx]*

All supplementary files are provided with the manuscript on submission. Raw FASTQ sequencing files are deposited at NCBI GEO under accession GSE[TBD] (to be populated at acceptance).

FIGURES

Figure 1. NES Heatmap — Key Pathway Enrichment Across 12 Comparisons

Normalized Enrichment Score (NES) from GSEAPreranked analysis. 42 curated pathways from four MSigDB databases across 12 pairwise comparisons (C1–C12). Color scale: red = positive NES (pathway enriched in numerator group); blue = negative NES (enriched in denominator); white = no enrichment. **Orange border** = statistically significant (|NES| > 1.5 and FDR < 0.25). Hover any cell for exact values.

−3 (neg) 0 +3 (pos) ★★ SIG: |NES|>1.5 and FDR<0.25 H Hallmark K KEGG R Reactome C CGP

Pathway	C1 AA vs NHW Par. Veh	C2 AA vs NHW Par. Ars	C3 NHW Ars vs Veh	C4 AA Ars vs Veh	C5 NHW Stem vs Par Veh	C6 AA Stem vs Par Veh	C7 NHW Stem vs Par Ars	C8 AA Stem vs Par Ars	C9 NHW Stem Ars vs Veh	C10 AA Stem Ars vs Veh	C11 AA vs NHW Stem Veh	C12 AA vs NHW Stem Ars
NER												
Nucleotide Excision Repair K	+0.98	+0.81	+1.10	+0.80	+0.81	+0.79	+0.86	-2.75	+1.08	+0.85	+0.63	-1.87
Nucleotide Excision Repair R	+0.99	+0.89	+1.21	+1.20	+1.01	+0.68	+1.14	-2.68	+1.36	+1.46	+0.72	-2.16
Global Genome Nucleotide Excision Repair Gg Ner R	+1.04	+0.92	+1.25	+1.24	+1.06	+0.78	+1.23	+0.67	+1.39	+1.53	+0.81	+0.62
Transcription Coupled Nucleotide Excision Repair Tc Ner R	+1.05	+0.94	+1.26	+1.25	+1.10	+0.80	+1.25	+0.74	+1.41	+1.56	+0.84	+0.65
Dual Incision In Gg Ner R	+1.15	+1.06	+1.41	+1.38	+1.27	+1.01	+1.55	+0.97	+1.58	+1.78	+1.01	+0.90
Formation Of Incision Complex In Gg Ner R	+1.27	+1.18	+1.56	+1.50	+1.50	+1.18	+1.89	+1.20	+1.73	+2.01	+1.17	+1.14
Formation Of Tc Ner Pre Incision Complex R	+1.15	+1.06	+1.41	+1.38	+1.27	+1.01	+1.55	+0.97	+1.58	+1.78	+1.01	+0.90
Newman Ercc6 Targets Dn C	+1.52	+1.43	+1.86	+1.70	+2.28	+1.77	+2.43	+1.91	+2.10	+2.35	+1.57	+1.61
BER												
Base Excision Repair R	+0.68	+0.60	+0.93	+0.89	-1.71	-1.73	-2.43	-1.67	+1.08	+1.02	-1.65	-1.42
Base Excision Repair Ap Site Formation R	+0.71	+0.63	+0.95	+0.91	-1.82	-1.77	+0.57	-1.71	+1.10	+1.04	-1.70	-1.45
MMR												
Mismatch Repair R	+0.61	-1.61	+0.88	+0.85	-1.60	-1.67	-1.76	-1.60	+1.03	+0.97	-1.56	-1.37
HR/DSB												
Homologous Recombination K	+0.58	-1.27	+0.79	-1.32	-1.50	-1.57	-1.61	-1.41	+0.88	-1.64	-1.25	-1.26
Hdr Through Homologous Recombination Hrr R	-0.98	-0.92	-1.24	-1.14	-1.07	-1.20	-1.19	-1.22	-1.75	-1.40	-1.05	-1.00
Diseases Of Dna Repair R	-1.23	-1.16	-1.79	-1.62	-1.29	-1.43	-1.42	-1.40	+0.84	+0.75	-1.29	-1.18
BROAD REPAIR												
Dna Repair H	+1.13	+0.96	+1.81	+1.40	+1.24	+1.18	+1.08	+1.38	+2.02	+1.59	+1.24	+1.29
EMT/STEMNESS												
Epithelial Mesenchymal Transition H	-1.22	-1.11	+0.99	-1.29	-1.34	-1.29	-1.26	-1.31	-1.48	-2.27	-1.42	-1.31
Foroutan Integrated Tgfb Emt Dn C	-1.67	-1.46	+1.00	-1.84	+0.94	+0.87	+0.87	+1.02	+0.92	+0.95	-1.95	-1.81
Benporath Es 1 C	-1.36	-1.27	-1.99	-1.41	-1.78	-1.65	-1.57	+0.73	-1.75	-1.97	-1.45	-1.42
Benporath Es 2 C	+1.32	+1.26	+1.66	+1.49	+1.84	+1.57	+1.93	+1.69	+1.72	+1.92	+1.41	+1.44
Lim Mammary Stem Cell Up C	-1.33	-1.25	-1.72	-1.36	-1.70	-1.59	-1.51	-2.25	-1.64	-1.79	-1.41	-1.39
Lim Mammary Luminal Progenitor Up C	-1.34	-1.26	-1.81	-1.38	-1.73	-1.61	-1.53	+0.60	-1.69	-1.85	-1.42	-1.41
Pece Mammary Stem Cell Up C	+1.32	+1.26	+1.66	+1.49	+1.85	+1.57	+1.94	+1.69	+1.72	+1.92	+1.41	+1.44
Boquest Stem Cell Up C	-1.42	-1.32	+0.80	-1.48	+0.52	-1.80	-1.68	+0.84	-1.96	+0.70	-1.53	-1.50
CELL CYCLE												
G2M Checkpoint H	-1.06	-0.97	-1.62	-1.10	-1.18	-1.13	-1.14	-1.15	-1.16	-1.07	-1.24	-1.21
E2F Targets H	-1.33	-1.16	+1.02	-1.37	-1.44	-1.33	-1.37	-1.45	-2.09	+0.83	-1.45	-1.36
Mitotic Spindle H	+0.88	-1.42	+1.26	+1.00	-1.72	-1.72	-1.61	-2.47	+1.14	+1.10	+0.97	+0.50
SIGNALING												
Hedgehog Signaling H	+1.28	+1.08	+2.00	+1.54	+1.97	+1.36	+2.02	+1.55	+2.15	+2.01	+1.46	+1.37
Signaling By Hedgehog R	+0.67	+0.59	+0.92	+0.89	-1.69	-1.72	-2.31	-1.64	+1.07	+1.01	-1.64	-1.41
Hedgehog Ligand Biogenesis R	+1.23	+1.13	+1.52	+1.46	+1.38	+1.10	+1.77	+1.11	+1.67	+1.90	+1.10	+1.07
Notch Signaling H	+1.42	+1.14	+2.06	+1.59	+2.40	+1.53	+2.75	+1.59	+2.43	+2.63	+1.54	+1.37
Tgf Beta Signaling H	+0.93	-1.51	+1.27	+1.03	-1.91	-2.27	-1.80	+0.94	+1.30	+1.18	+1.04	+0.91
Wnt Beta Catenin Signaling H	+1.04	+0.77	+1.43	+1.23	+1.02	+0.95	-1.97	+1.13	+1.56	+1.41	+1.10	+1.09
Myc Targets V1 H	+1.14	+0.98	+1.83	+1.40	+1.40	+1.20	+1.14	+1.42	+2.05	+1.66	+1.33	+1.31
Myc Targets V2 H	+0.79	-1.31	+1.21	+0.93	-1.67	-1.58	-1.52	-1.88	+1.10	+1.01	+0.70	-2.05
STRESS/ARSENIC												
Inflammatory Response H	-1.69	-1.31	+1.15	+0.92	-1.63	-1.56	-1.45	-1.66	+1.03	+0.99	-2.30	-1.99
Oxidative Phosphorylation H	+1.68	+1.26	+2.14	+1.83	+2.48	+1.68	+2.76	+1.61	+2.51	+2.67	+1.77	+1.72
Tnfa Signaling Via Nfkb H	-1.57	-1.29	+1.15	+0.86	-1.61	-1.48	-1.45	-1.62	+1.01	+0.96	-2.23	-1.99
Xenobiotic Metabolism H	-1.38	-1.19	+1.06	-1.50	-1.56	-1.36	-1.37	-1.49	+0.84	+0.85	-1.82	-1.42
Hypoxia H	+1.32	+1.10	+2.05	+1.57	+2.31	+1.45	+2.49	+1.58	+2.19	+2.35	+1.51	+1.37
Uv Response Dn H	-1.53	-1.22	+1.12	+0.75	-1.58	-1.47	-1.43	-1.59	+1.01	+0.94	-1.89	-1.53
Nuclear Events Mediated By Nfe2L2 R	-1.18	-1.11	-1.65	-1.50	-1.25	-1.37	-1.37	-1.37	+0.81	+0.67	-1.22	-1.15
Kan Response To Arsenic Trioxide C	-1.54	-1.39	+0.93	-1.63	+0.86	+0.71	+0.73	+0.96	+0.80	+0.87	-1.69	-1.65

Figure 1. NES Heatmap — Key Pathway Enrichment Across 12 Comparisons.

Normalized Enrichment Score (NES) from GSEAPreranked analysis of 42 curated pathways across 12 pairwise comparisons (C1–C12) spanning parental and CD24−/CD49f+/CD44+ sorted stem cell populations under vehicle and arsenic trioxide conditions (0.3 µM, 14 days) in BRL-23 (NHW) and BRL-24 (AA) breast tissue-derived cell lines. Each cell displays the NES value for a given pathway in a given comparison; red = positive NES (enriched in numerator group); blue = negative NES (enriched in denominator); white = no enrichment. Color scale clamped at ±3. Pathways are grouped by functional category (NER, BER, MMR, HR/DSB, EMT/Stemness, Cell Cycle, Signaling, Stress/Arsenic). Comparison columns are color-coded by type:

racial comparisons (C1, C2, C11, C12), NHW arsenic comparisons (C3, C7), AA arsenic comparisons (C4, C8), NHW stem comparisons (C5, C9), AA stem comparisons (C6, C10). Orange cell border = statistically significant (|NES| > 1.5 and FDR < 0.25). Gene set database source: H = Hallmark, K = KEGG, R = Reactome, C = CGP. The primary finding — selective suppression of KEGG_NUCLEOTIDE_EXCISION_REPAIR in C8 (AA Stem vs. Parental Arsenic; NES = −2.75, FDR = 0.18) — is the most deeply blue bordered cell, with no equivalent suppression in C7 (NHW Stem vs. Parental Arsenic).

Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **202 up** / **231 dn** NER sig: **7** Stemness sig: **12**

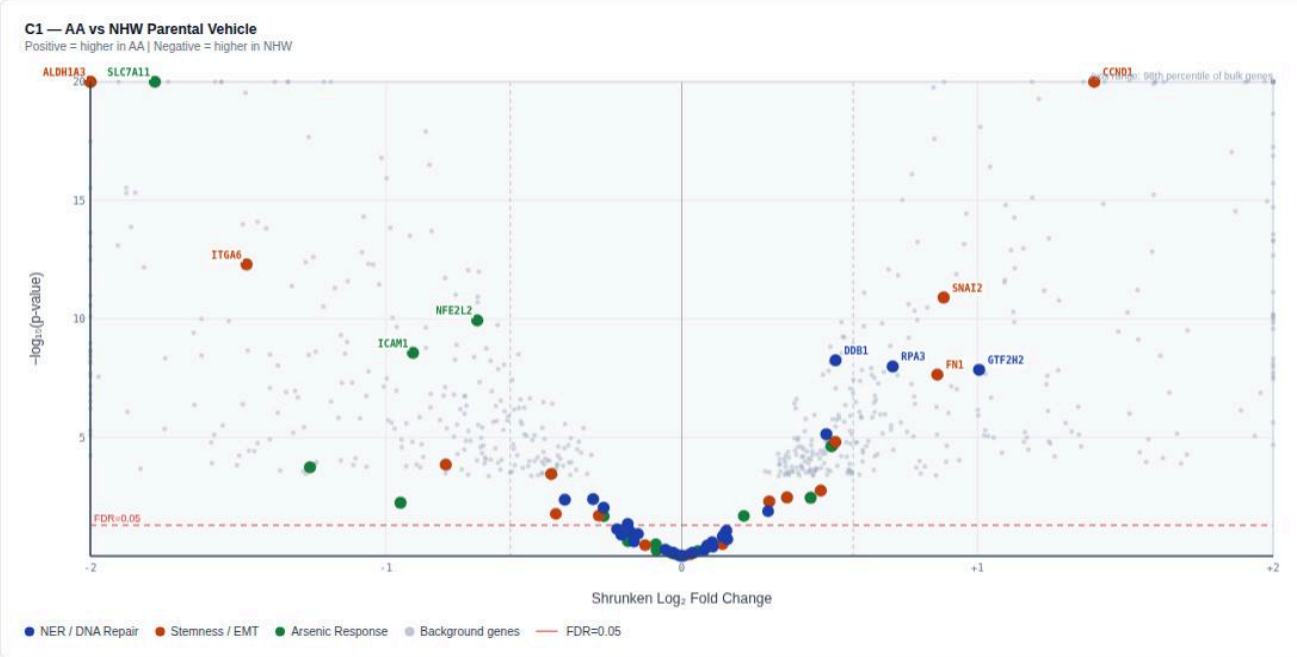


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 | **C2** | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12

FDR<0.05, |L2FC|>0.58: **181 up** / **286 dn** NER sig: **12** Stemness sig: **13**

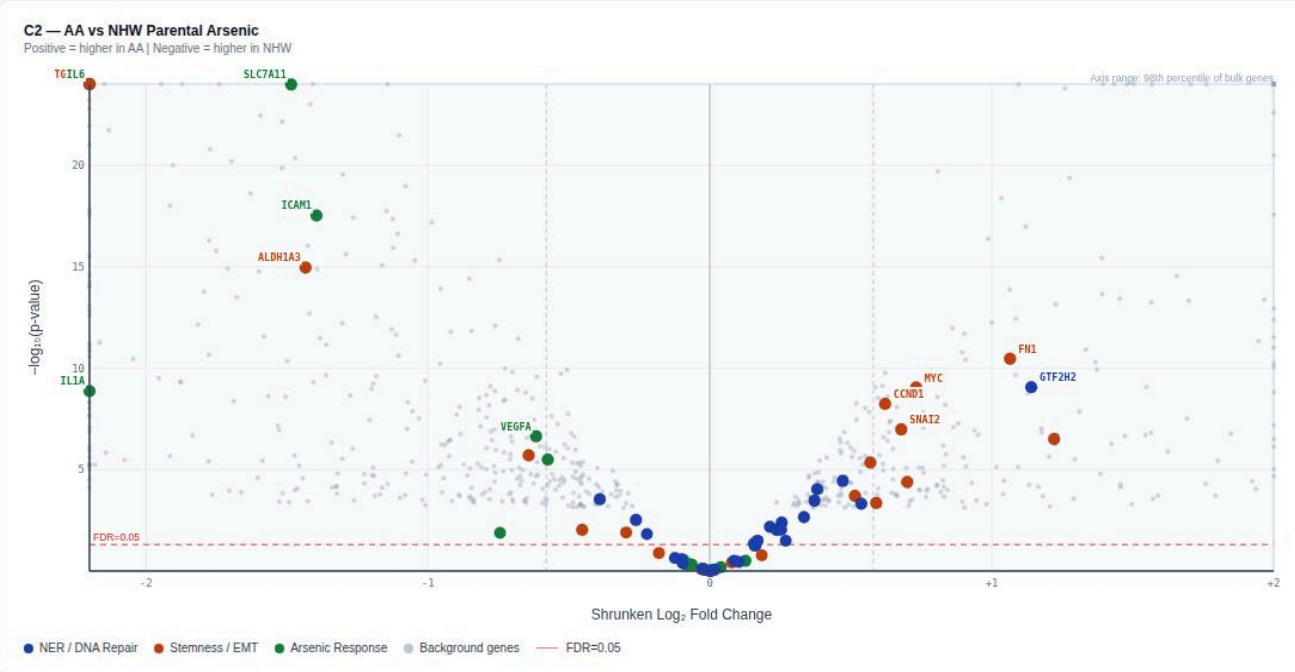


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 | C2 | **C3** | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12

FDR<0.05, |L2FC|>0.58: **164 up** / **205 dn** NER sig: **7** Stemness sig: **8**

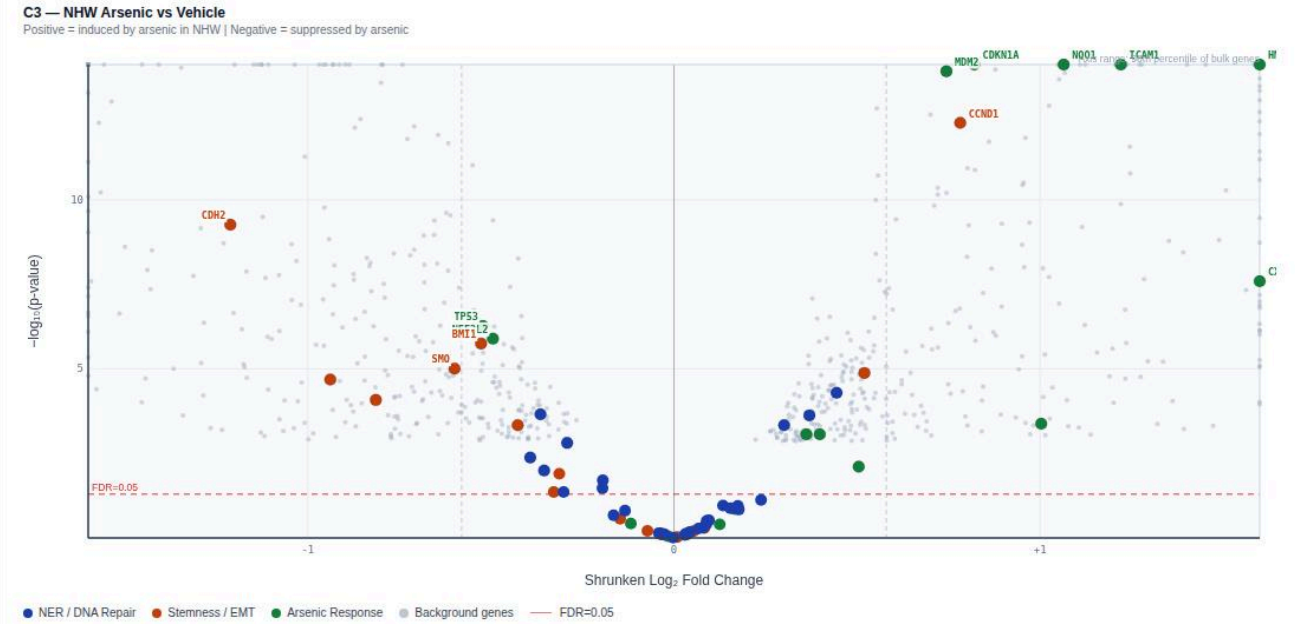


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 **C4** C5 C6 C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **137** up / **202** dn NER sig: **9** Stemness sig: **7**

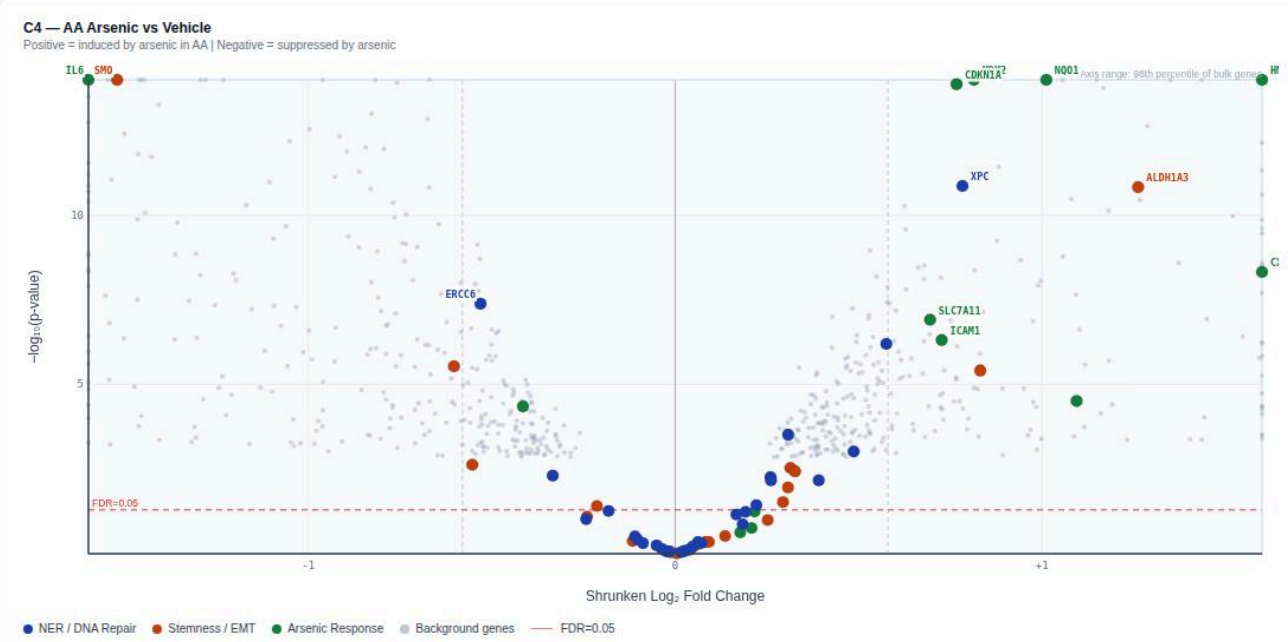


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 **C5** C6 C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **266** up / **179** dn NER sig: **9** Stemness sig: **7**

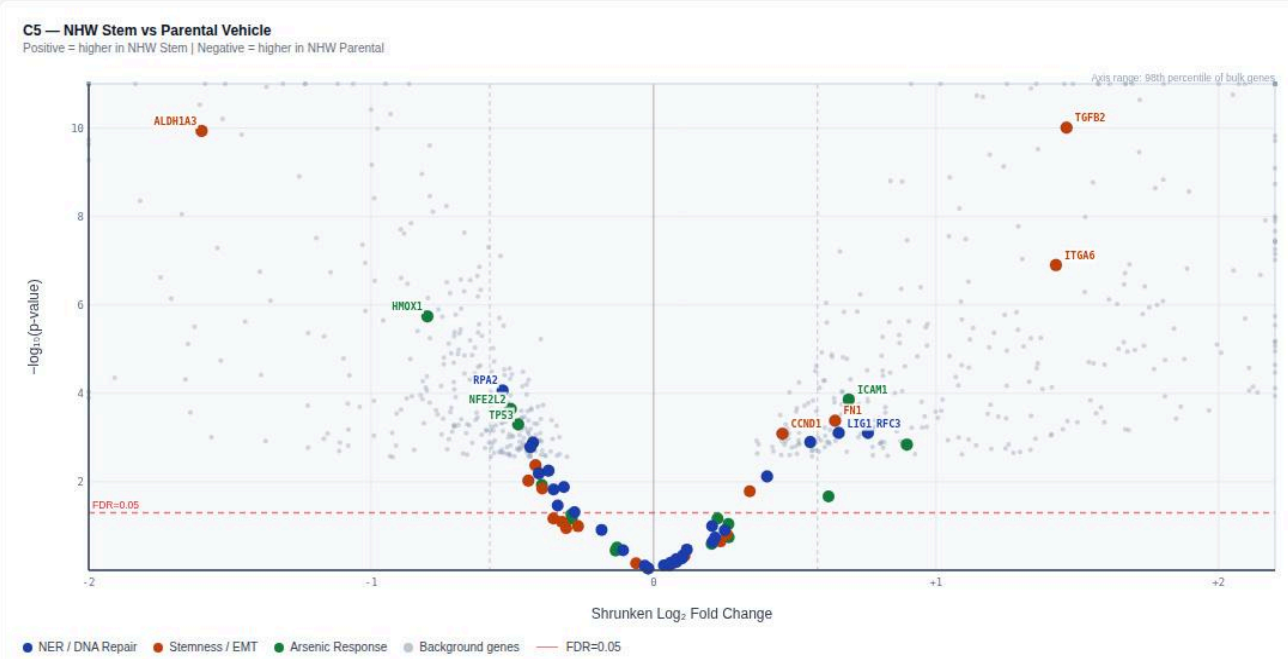


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 **C6** C7 C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **238 up** / **227 dn** NER sig: **6** Stemness sig: **4**

C6 — AA Stem vs Parental Vehicle

Positive = higher in AA Stem | Negative = higher in AA Parental

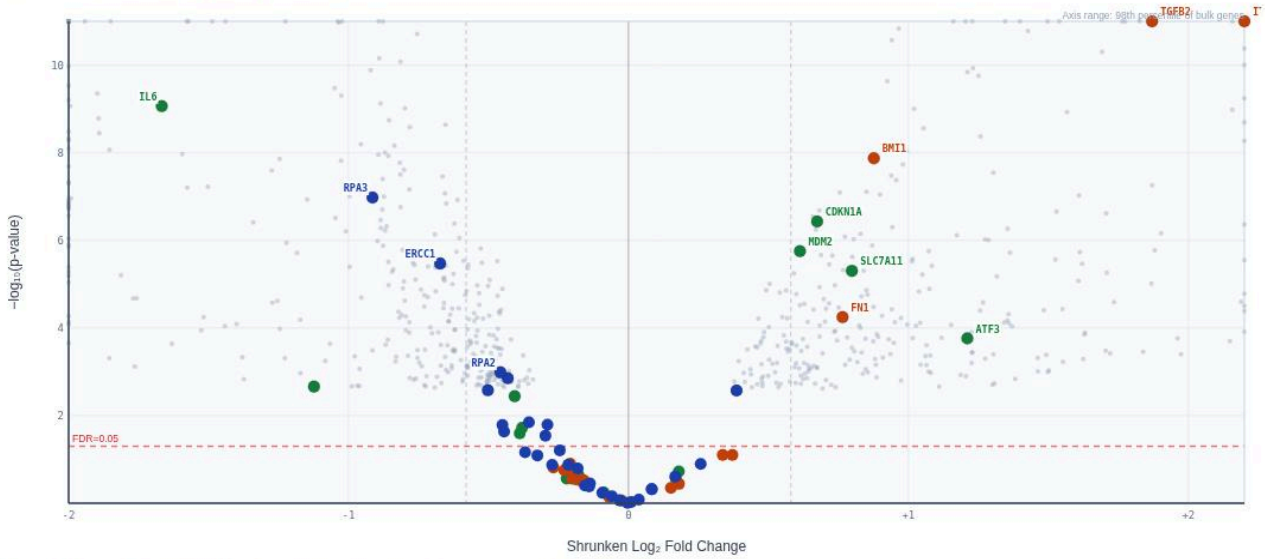


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 **C7** C8 C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **263 up** / **196 dn** NER sig: **8** Stemness sig: **7**

C7 — NHW Stem vs Parental Arsenic

Positive = higher in NHW Stem | Negative = higher in NHW Parental Ars

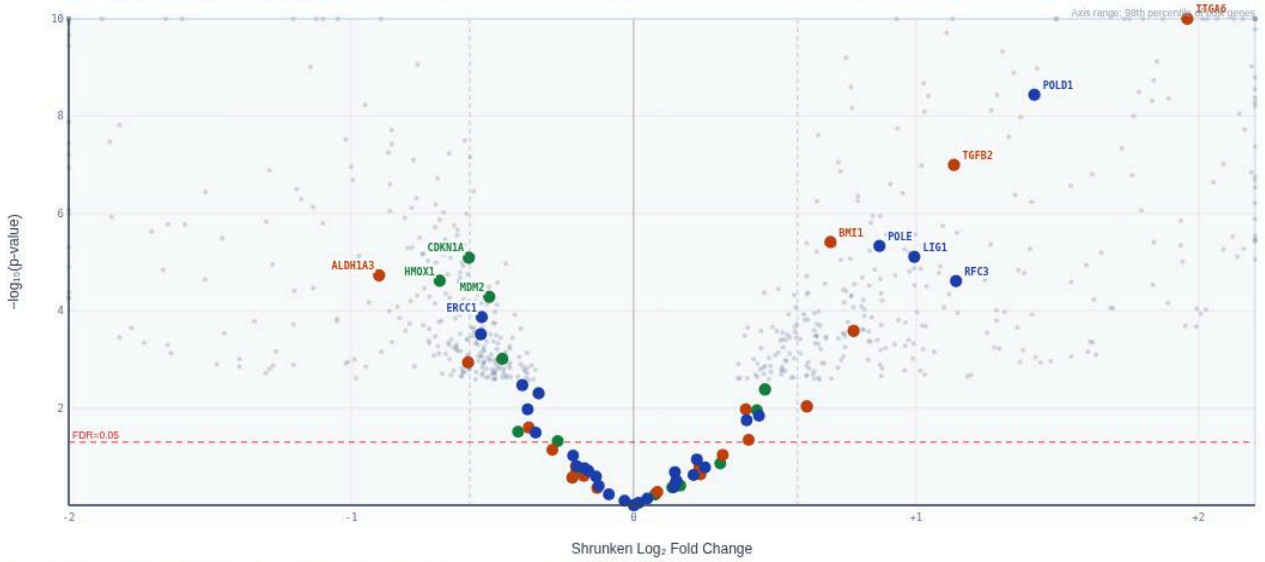


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 **C8** C9 C10 C11 C12

FDR<0.05, |L2FC|>0.58: **216 up** / **184 dn** NER sig: **12** Stemness sig: **6**

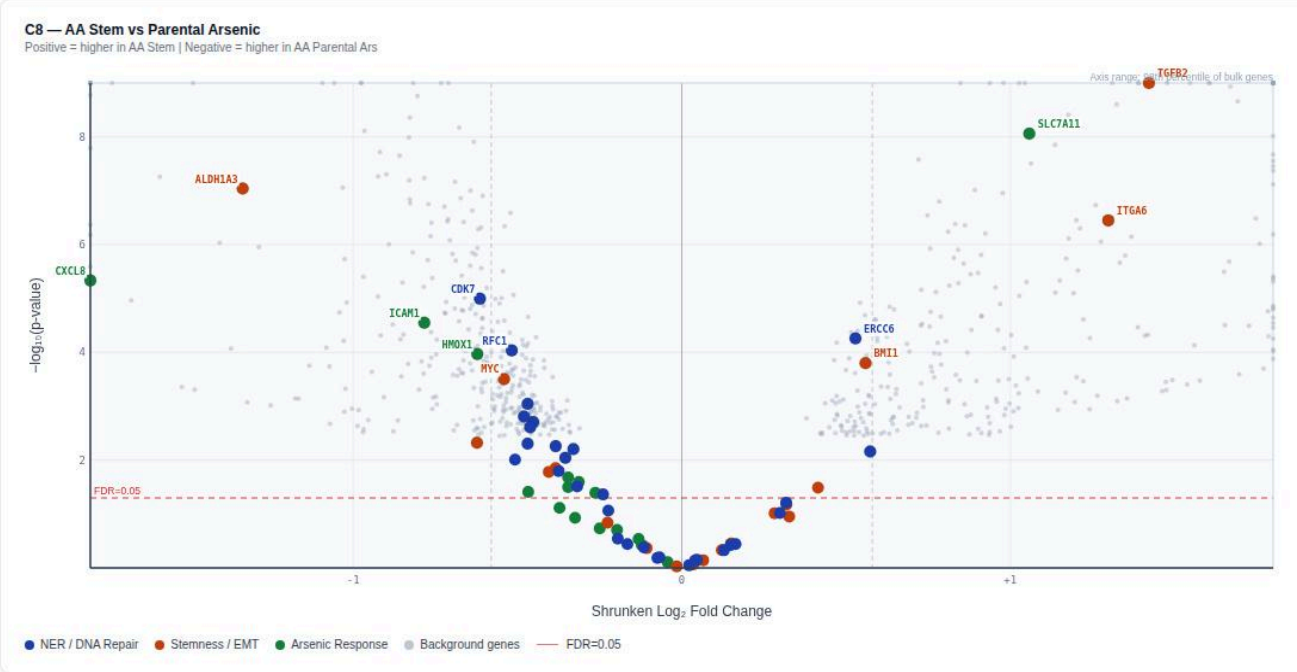


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 **C9** C10 C11 C12

FDR<0.05, |L2FC|>0.58: **40 up** / **45 dn** NER sig: **1** Stemness sig: **2**

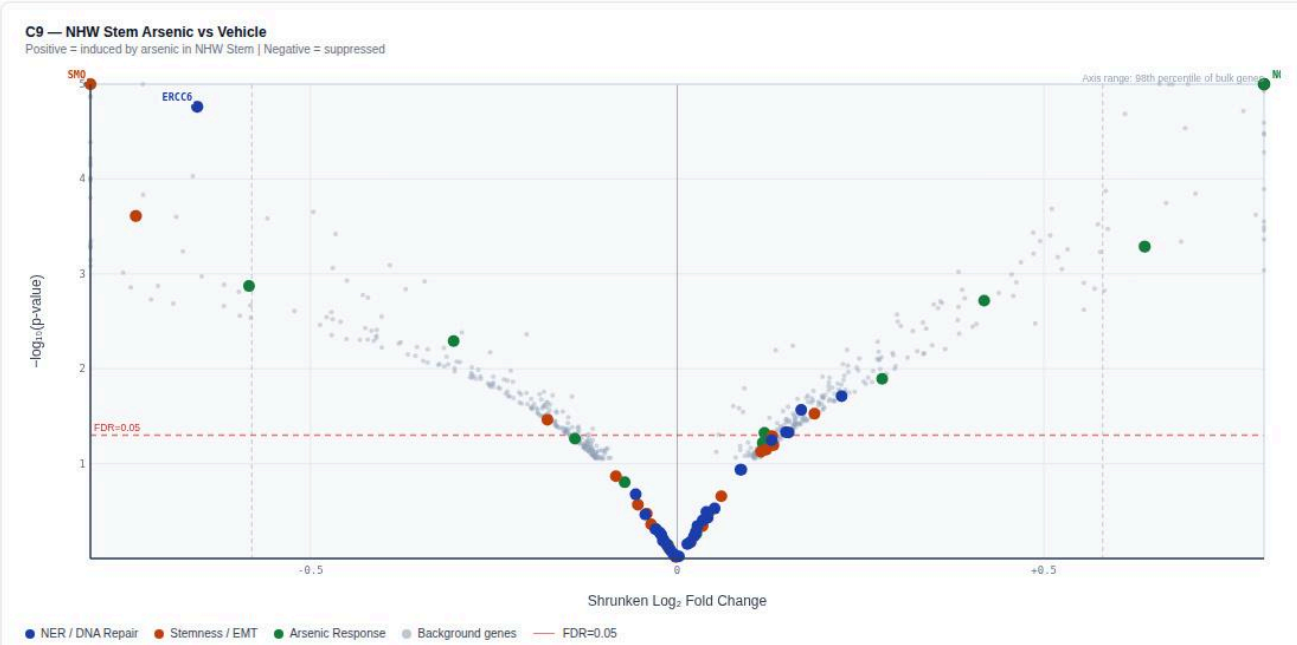


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 **C10** C11 C12

FDR<0.05, |L2FC|>0.58: **47 up / 30 dn** NER sig: **0** Stemness sig: **1**

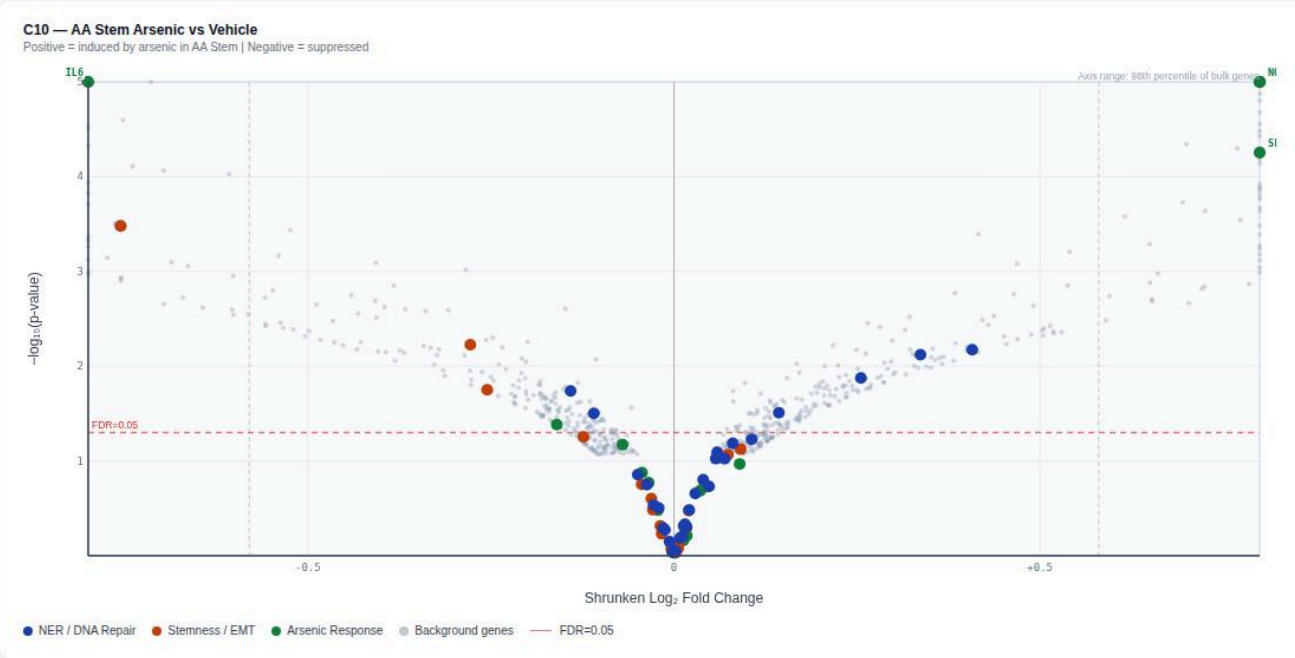


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 **C11** C12

FDR<0.05, |L2FC|>0.58: **200 up** / **316 dn** NER sig: **8** Stemness sig: **6**

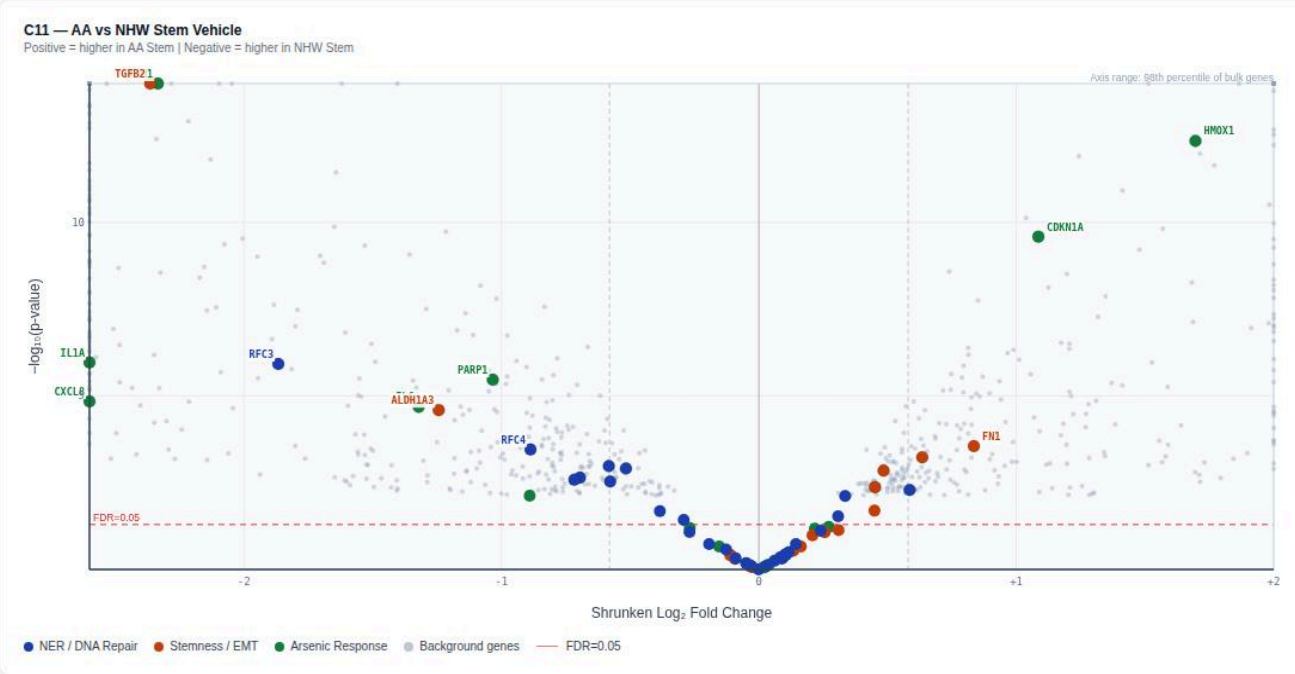


Figure 2. Volcano Plots — Differential Expression (All 12 Comparisons)

Shrunken Log_2 Fold Change (DESeq2 apeglm) vs $-log_{10}$(p-value). Axes zoomed to 98th percentile of bulk genes to show the relevant range clearly. NER genes: blue · Stemness/EMT: orange · Arsenic response: green · Background: gray. Dashed red line = FDR 0.05 threshold. Vertical dashes = ±0.58 (1.5-fold). Hover for gene details.

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 **C12**

FDR<0.05, |L2FC|>0.58: **198 up** / **243 dn** NER sig: **2** Stemness sig: **3**

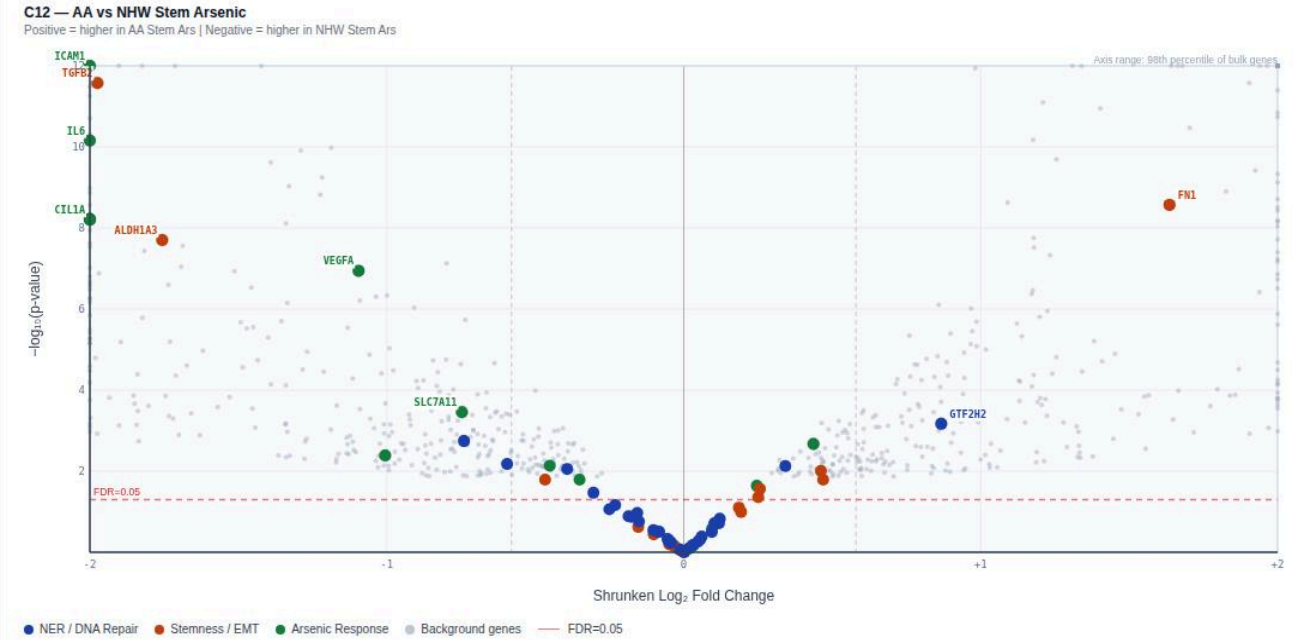


Figure 2. Volcano Plots — Differential Gene Expression Across All 12 Comparisons (Panels A–L).

Volcano plots displaying DESeq2 differential expression results for each pairwise comparison: (A) C1, AA vs. NHW Parental Vehicle; (B) C2, AA vs. NHW Parental Arsenic; (C) C3, NHW Arsenic vs. Vehicle; (D) C4, AA Arsenic vs. Vehicle; (E) C5, NHW Stem vs. Parental Vehicle; (F) C6, AA Stem vs. Parental Vehicle; (G) C7, NHW Stem vs. Parental Arsenic; (H) C8, AA Stem vs. Parental Arsenic; (I) C9, NHW Stem Arsenic vs. Vehicle; (J) C10, AA Stem Arsenic vs. Vehicle; (K) C11, AA vs. NHW Stem Vehicle; (L) C12, AA vs. NHW Stem Arsenic. X-axis: shrunken $\log_2$ fold change (DESeq2 apeglm); y-axis: $-\log_{10}$(p-value). Axes zoomed to the 98th percentile of background genes to display the biologically relevant range clearly. Colored points: blue = NER/DNA repair genes; orange/red = Stemness/EMT genes; green = arsenic response genes; gray = all other genes. Horizontal red dashed line = FDR 0.05. Vertical dashed lines = ±0.58 $\log_2$ FC (1.5-fold). Selected genes are labeled. Gene counts above each panel report significant DEGs (FDR < 0.05, |L2FC| > 0.58). Panels I and J (C9 and C10, n=1 per group) display a compressed axis range reflecting reduced statistical power due to single-replicate stem cell conditions.

Figure 3. Individual Gene Expression — Shrunken Log_2 Fold Change (All 12 Comparisons)

Shrunken Log_2 fold change (DESeq2 apeglm) for key genes across all 12 pairwise comparisons C1–C12. Bars colored by gene category; faded bars = FDR≥0.05 (not significant). Asterisks: * FDR<0.05 ** FDR<0.01 *** FDR<0.001. Bar width scaled within each gene card to its own maximum observed L2FC.

Category: All Genes | NER / DNA Repair | Stemness / EMT | Arsenic Response

NER / DNA Repair (solid = sig) Stemness / EMT Arsenic Response ★ FDR<0.05 ★★ FDR<0.01 ★★★ FDR<0.001

Figure 3. Individual Gene Expression — Shrunken Log_2 Fold Change (All 12 Comparisons)

Shrunken Log_2 fold change (DESeq2 apeglm) for key genes across all 12 pairwise comparisons C1–C12. Bars colored by gene category; faded bars = FDR≥0.05 (not significant). Asterisks: * FDR<0.05 ** FDR<0.01 *** FDR<0.001. Bar width scaled within each gene card to its own maximum observed L2FC.

Category: All Genes | NER / DNA Repair | Stemness / EMT | Arsenic Response

NER / DNA Repair (solid = sig) Stemness / EMT Arsenic Response ★ FDR<0.05 ★★ FDR<0.01 ★★★ FDR<0.001

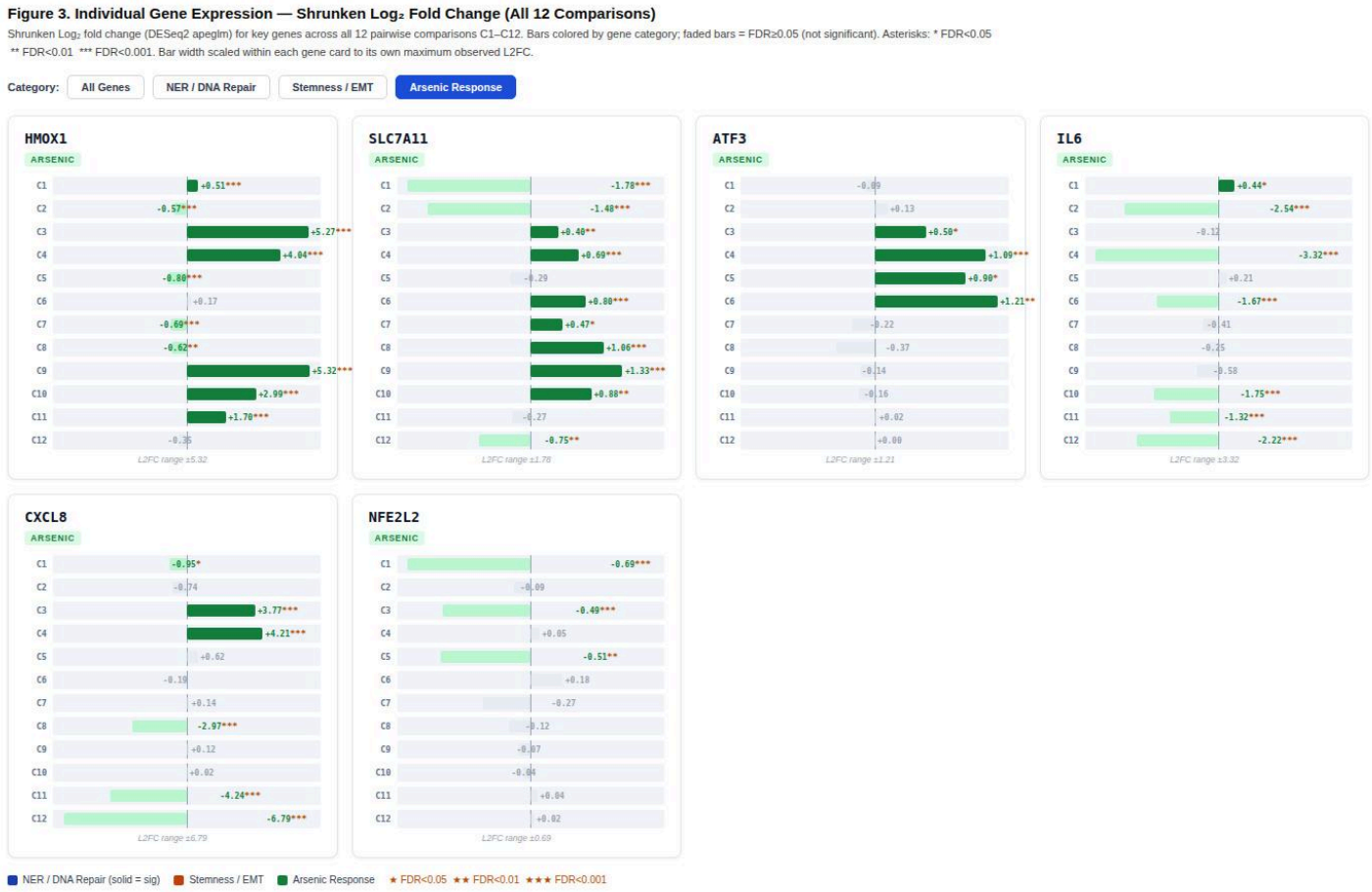


Figure 3. Individual Gene Expression — Shrunken Log$_2$ Fold Change Across All 12 Comparisons (Panels A–C).

Bar charts displaying the shrunken log$_2$ fold change (DESeq2 apeglm) for key individual genes across all 12 pairwise comparisons (C1–C12), organized by gene category: (A) NER / DNA repair genes: ERCC1, XPC, DDB1, DDB2, ERCC3, ERCC6, CDK7, CETN2, RAD23A, POLD1, LIG1, and PCNA. (B) Stemness / EMT genes: SNAI2, VIM, KLF4, BMI1, SMO, ALDH1A3, CD44, and TGFBR1. (C) Arsenic response genes: HMOX1, SLC7A11, ATF3, IL6, CXCL8, and NFE2L2. Bar color indicates gene category (blue = NER/DNA repair; orange/red = Stemness/EMT; green = Arsenic response). Solid bars = FDR < 0.05 (significant); faded bars = FDR ≥ 0.05. Bar width is scaled within each gene card to its own maximum observed L2FC. Significance asterisks: * FDR < 0.05; ** FDR < 0.01; *** FDR < 0.001. Positive L2FC = higher expression in the numerator group; negative L2FC = higher expression in the denominator. ERCC1 (panel A) shows consistent suppression specifically in sorted-stem-vs-parental comparisons in both racial lines under arsenic (C7, C8) and in the racial stem-arsenic contrast (C12), establishing it as the most reproducible single-gene marker of the racial NER disparity.

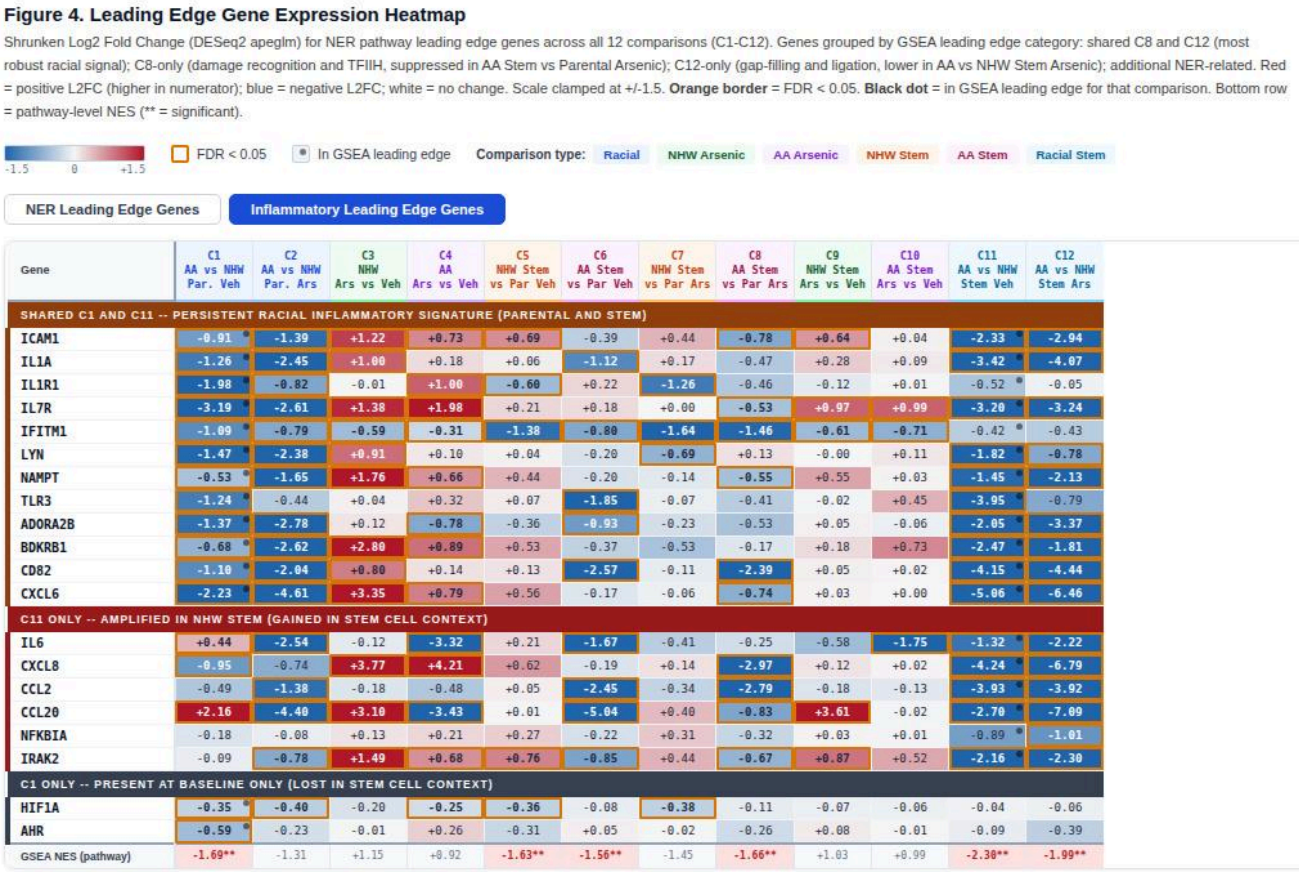
Figure 4. Leading Edge Gene Expression Heatmap

Shrunken Log2 Fold Change (DESeq2 apeglm) for NER pathway leading edge genes across all 12 comparisons (C1-C12). Genes grouped by GSEA leading edge category: shared C8 and C12 (most robust racial signal); C8-only (damage recognition and TFIIH, suppressed in AA Stem vs Parental Arsenic); C12-only (gap-filling and ligation, lower in AA vs NHW Stem Arsenic); additional NER-related. Red = positive L2FC (higher in numerator); blue = negative L2FC; white = no change. Scale clamped at +/-1.5. **Orange border** = FDR < 0.05. **Black dot** = in GSEA leading edge for that comparison. Bottom row = pathway-level NES (** = significant).

-1.5 0 +1.5 | FDR < 0.05 | In GSEA leading edge | **Comparison type:** Racial | NHW Arsenic | AA Arsenic | NHW Stem | AA Stem | Racial Stem

NER Leading Edge Genes | Inflammatory Leading Edge Genes

Gene	C1 AA vs NHW Par. Veh	C2 AA vs NHW Par. Ars	C3 NHW Ars vs Veh	C4 AA Ars vs Veh	C5 NHW Stem vs Par Veh	C6 AA Stem vs Par Veh	C7 NHW Stem vs Par Ars	C8 AA Stem vs Par Ars	C9 NHW Stem Ars vs Veh	C10 AA Stem Ars vs Veh	C11 AA vs NHW Stem Veh	C12 AA vs NHW Stem Ars
SHARED C8 AND C12 -- MOST ROBUST RACIAL NER SIGNAL												
CCNH	-0.18	-0.39	+0.44	+0.22	+0.06	-0.14	-0.38	-0.48 •	-0.00	-0.01	-0.29	-0.39 •
ERCC1	+0.14	-0.22	+0.37	-0.02	-0.44	-0.67	-0.54	-0.47 •	+0.09	+0.02	-0.05	-0.08 •
RFC2	-0.20	+0.10	+0.04	+0.39	+0.05	-0.44	+0.25	-0.51 •	+0.05	+0.05	-0.72	-0.59 •
RPA3	+0.71	+0.33	+0.17	-0.11	+0.40	-0.91	+0.15	-0.37 •	-0.01	+0.08	-0.58	-0.09 •
C8 ONLY -- DAMAGE RECOGNITION AND TFIIH LOADING (SUPPRESSED IN AA STEM VS PARENTAL ARSENIC)												
XPC	-0.30	+0.37	+0.03	+0.78	-0.28	+0.17	-0.20	-0.45 •	+0.03	+0.01	+0.11	+0.06
DDB1	+0.52	+0.17	+0.30	-0.02	-0.18	-0.29	-0.34	-0.33 •	+0.04	-0.01	+0.34	+0.10
ERCC3	-0.00	+0.16	+0.13	+0.31	-0.11	+0.08	-0.13	-0.24 •	+0.03	-0.00	+0.14	+0.03
GTF2H1	+0.04	+0.02	+0.09	+0.07	-0.35	-0.00	-0.54	-0.32 •	-0.02	-0.04	+0.31	+0.12
CDK7	+0.14	+0.09	-0.13	-0.18	-0.43	-0.43	-0.39	-0.61 •	-0.02	-0.11	+0.10	-0.05
CETN2	+0.29	+0.00	-0.04	-0.33	-0.34	-0.50	-0.35	-0.47 •	-0.01	-0.05	+0.09	-0.05
RAD23A	-0.02	+0.47	+0.07	+0.58	-0.41	-0.06	-0.12	-0.46 •	+0.13	+0.02	+0.24	+0.05
RFC1	-0.18	+0.16	-0.20	+0.17	-0.32	-0.18	-0.21	-0.52 •	-0.02	-0.02	-0.03	-0.05
RPA2	-0.15	+0.24	-0.36	+0.05	-0.53	-0.46	-0.03	-0.38 •	+0.03	+0.01	-0.05	-0.04
POLD2	+0.09	+0.38	-0.03	+0.26	-0.37	-0.30	+0.02	-0.35 •	+0.17	+0.03	+0.12	+0.00
POLE3	+0.01	-0.07	-0.01	-0.10	-0.41	-0.48	-0.41	-0.51 •	-0.00	-0.01	-0.04	-0.08
CUL4A	+0.17	-0.01	+0.42	+0.23	-0.07	-0.20	-0.27	-0.23 •	+0.06	+0.03	+0.03	+0.02
C12 ONLY -- GAP-FILLING AND LIGATION (LOWER IN AA VS NHW STEM ARSENIC)												
PCNA	-0.26	-0.10	+0.04	+0.19	+0.08	-0.21	+0.05	-0.11	+0.00	+0.06	-0.52	-0.16 •
LIG1	+0.08	+0.27	+0.24	+0.49	+0.66	-0.16	+0.99	-0.17	+0.15	+0.11	-0.69	-0.74 •
DDB2	+0.04	-0.12	+0.08	-0.10	+0.22	+0.04	+0.00	-0.07	-0.03	-0.03	-0.09	-0.10 •
GTF2H4	+0.01	+0.00	+0.03	+0.03	-0.02	-0.33	+0.14	-0.19	+0.04	+0.02	-0.19	-0.19 •
CUL4B	-0.85	-0.91	-0.17	-0.20	-0.09	-0.08	-0.32	+0.07	-0.12	-0.00	-0.63	-0.16 •
RFC3	-0.16	-0.10	-0.30	-0.24	+0.76	-0.37	+1.14	+0.57	-0.02	+0.41	-1.87	-0.25 •
RFC4	-0.40	-0.02	-0.39	-0.02	-0.02	-0.45	+0.44	+0.13	+0.02	+0.26	-0.89	-0.18 •
RFC5	-0.22	-0.00	-0.03	+0.18	+0.25	-0.14	+0.40	+0.02	+0.02	+0.07	-0.58	-0.23 •
RPA4	+0.06	-0.08	+0.26	+0.07	+0.52	+0.07	+0.33	-0.11	+0.02	-0.02	-0.18	-0.57 •
POLD1	+0.10	+0.54	-0.35	+0.02	+0.21	-0.27	+1.42	+0.30	+0.22	+0.34	-0.27	-0.15 •
POLD3	-0.47	+0.10	-0.51	+0.09	+0.04	-0.37	+0.73	-0.17	+0.04	+0.05	-0.96	-0.79 •
POLE	+0.15	+0.01	+0.18	+0.04	+0.55	-0.09	+0.87	+0.32	+0.15	+0.14	-0.38	-0.30 •
POLE2	-0.29	-0.07	-0.19	-0.01	+0.26	-0.30	+0.75	+0.28	+0.01	+0.21	-1.87	-0.18 •
ADDITIONAL NER-RELATED												
PARP1	-0.26	-0.02	-0.02	+0.22	+0.26	-0.38	+0.15	-0.35	-0.03	+0.04	-1.03	-0.45
PARP2	-0.21	+0.35	-0.14	+0.46	-0.10	-0.01	+0.05	-0.59	+0.01	-0.01	-0.07	-0.14
ERCC6	-0.06	-0.26	-0.29	-0.53	+0.21	+0.39	-0.17	+0.53	-0.65	-0.14	+0.08	+0.34
GSEA NES (pathway)	+1.68	+1.26	+1.86	+1.70	+2.28	+1.77	+2.43	-2.75**	+2.10	+2.35	+1.57	-1.87**



Gene	C1 AA vs NHW Par. Veh	C2 AA vs NHW Par. Ars	C3 NHW Ars vs Veh	C4 AA Ars vs Veh	C5 NHW Stem vs Par Veh	C6 AA Stem vs Par Veh	C7 NHW Stem vs Par Ars	C8 AA Stem vs Par Ars	C9 NHW Stem Ars vs Veh	C10 AA Stem Ars vs Veh	C11 AA vs NHW Stem Veh	C12 AA vs NHW Stem Ars
SHARED C1 AND C11 -- PERSISTENT RACIAL INFLAMMATORY SIGNATURE (PARENTAL AND STEM)												
ICAM1	-0.91	-1.39	+1.22	+0.73	+0.69	-0.39	+0.44	-0.78	+0.64	+0.04	-2.33	-2.94
IL1A	-1.26	-2.45	+1.00	+0.18	+0.06	-1.12	+0.17	-0.47	+0.28	+0.09	-3.42	-4.07
IL1R1	-1.98	-0.82	-0.01	+1.00	-0.60	+0.22	-1.26	-0.46	-0.12	+0.01	-0.52	-0.05
IL7R	-3.19	-2.61	+1.38	+1.98	+0.21	+0.18	+0.00	-0.53	+0.97	+0.99	-3.20	-3.24
IFITM1	-1.09	-0.79	-0.59	-0.31	-1.38	-0.80	-1.64	-1.46	-0.61	-0.71	-0.42	-0.43
LYN	-1.47	-2.38	+0.91	+0.10	+0.04	-0.20	-0.69	+0.13	-0.00	+0.11	-1.82	-0.78
NAMPT	-0.53	-1.65	+1.76	+0.66	+0.44	-0.20	-0.14	-0.55	+0.55	+0.03	-1.45	-2.13
TLR3	-1.24	-0.44	+0.04	+0.32	+0.07	-1.85	-0.07	-0.41	-0.02	+0.45	-3.95	-0.79
ADORA2B	-1.37	-2.78	+0.12	-0.78	-0.36	-0.93	-0.23	-0.53	+0.05	-0.06	-2.05	-3.37
BDKRB1	-0.68	-2.62	+2.80	+0.89	+0.53	-0.37	-0.53	-0.17	+0.18	+0.73	-2.47	-1.81
CD82	-1.10	-2.04	+0.80	+0.14	+0.13	-2.57	-0.11	-2.39	+0.05	+0.02	-4.15	-4.44
CXCL6	-2.23	-4.61	+3.35	+0.79	+0.56	-0.17	-0.06	-0.74	+0.03	+0.00	-5.06	-6.46
C11 ONLY -- AMPLIFIED IN NHW STEM (GAINED IN STEM CELL CONTEXT)												
IL6	+0.44	-2.54	-0.12	-3.32	+0.21	-1.67	-0.41	-0.25	-0.58	-1.75	-1.32	-2.22
CXCL8	-0.95	-0.74	+3.77	+4.21	+0.62	-0.19	+0.14	-2.97	+0.12	+0.02	-4.24	-6.79
CCL2	-0.49	-1.38	-0.18	-0.48	+0.05	-2.45	-0.34	-2.79	-0.18	-0.13	-3.93	-3.92
CCL20	+2.16	-4.40	+3.10	-3.43	+0.01	-5.04	+0.40	-0.83	+3.61	-0.02	-2.70	-7.09
NFKBIA	-0.18	-0.08	+0.13	+0.21	+0.27	-0.22	+0.31	-0.32	+0.03	+0.01	-0.89	-1.01
IRAK2	-0.09	-0.78	+1.49	+0.68	+0.76	-0.85	+0.44	-0.67	+0.87	+0.52	-2.16	-2.30
C1 ONLY -- PRESENT AT BASELINE ONLY (LOST IN STEM CELL CONTEXT)												
HIF1A	-0.35	-0.40	-0.20	-0.25	-0.36	-0.08	-0.38	-0.11	-0.07	-0.06	-0.04	-0.06
AHR	-0.59	-0.23	-0.01	+0.26	-0.31	+0.05	-0.02	-0.26	+0.08	-0.01	-0.09	-0.39
GSEA NES (pathway)	-1.69**	-1.31	+1.15	+0.92	-1.63**	-1.56**	-1.45	-1.66**	+1.03	+0.99	-2.30**	-1.99**

Figure 4. Leading Edge Gene Expression Heatmaps (Panels A–B).

Shrunken $\log_2$ fold change (DESeq2 apeglm) for GSEA leading edge genes across all 12 comparisons (C1–C12). Red = positive L2FC (higher in numerator); blue = negative L2FC (higher in denominator); white = no change. Scale clamped at ±1.5. Orange border = FDR < 0.05. Black dot = gene is in the GSEA leading edge for that comparison. Bottom row = pathway-level NES (** = |NES| > 1.5 and FDR < 0.25). Comparison columns are color-coded by type. Row groupings within each panel are indicated by colored row-bracket annotations on the left margin of the heatmap, separating the Shared, group-specific, and contextual categories described below. (A) NER pathway leading edge genes, organized into four groups — Shared C8 and C12: genes present in both the C8 (AA Stem vs. Parental Arsenic) and C12 (AA vs. NHW Stem Arsenic) KEGG NER leading edges, representing the most reproducible racial NER markers (CCNH, ERCC1, RFC2, RPA3); C8-only: damage recognition and TFIIH loading factors suppressed in AA stem cells under arsenic (XPC, DDB1, ERCC3, GTF2H1, CDK7, CETN2, RAD23A, RFC1, RPA2, POLD2, POLE3, CUL4A); C12-only: gap-filling and ligation machinery lower in AA vs. NHW stem cells under arsenic (PCNA, LIG1, DDB2, GTF2H4, CUL4B, RFC3, RFC4, RFC5, RPA4, POLD1, POLD3, POLE, POLE2); additional NER-related genes shown for context (PARP1, PARP2, ERCC6). (B) Inflammatory pathway leading edge genes, organized into three groups — Shared C1 and C11: genes in the leading edge of both the parental (C1) and stem (C11) racial inflammatory comparisons, representing the persistent cell-intrinsic NHW inflammatory signature (ICAM1, IL1A, IL1R1, IL7R, IFITM1, LYN, NAMPT, TLR3, ADORA2B, BDKRB1, CD82, CXCL6); C11-only: genes gained exclusively in the NHW stem cell inflammatory leading edge, representing the stemness-amplified inflammatory program (IL6, CXCL8, CCL2, CCL20, NFKBIA, IRAK2); C1-only: genes present at baseline only, absent in stem cell context (HIF1A, AHR).

REFERENCES

1. Latimer JJ, Johnson JM, Miles TD, et al. Cell-type-specific level of DNA nucleotide excision repair in primary cultures of human mammary epithelial cells versus fibroblasts. DNA Repair. 2010;9(1):30-37. doi:10.1016/j.dnarep.2009.10.004

2. Latimer JJ. Culture media for non-diseased human epithelial cells. US Patent No. 6,383,805B1. 2002.

3. Alharbi YN. Analysis of pro-carcinogenic exposures that impact DNA repair gene expression from arsenic trioxide, glyphosate and the Gulf War with a focus on ancestral disparity [Dissertation]. Nova Southeastern University; 2025.

4. Environmental Working Group. EWG Tap Water Database. https://www.ewg.org/tapwater. Accessed 2024.

5. Shen ZX, Chen GQ, Ni JH, et al. Use of arsenic trioxide in the treatment of acute promyelocytic leukemia: clinical efficacy and pharmacokinetics in relapsed patients. Blood. 1997;89(9):3354-3360.

6. Al-Hajj M, Wicha MS, Benito-Hernandez A, Morrison SJ, Clarke MF. Prospective identification of tumorigenic breast cancer cells. Proc Natl Acad Sci USA. 2003;100(7):3983-3988. doi:10.1073/pnas.0530291100

7. Shackleton M, Vaillant F, Simpson KJ, et al. Generation of a functional mammary gland from a single stem cell. Nature. 2006;439(7072):84-88. doi:10.1038/nature04372

8. Pham PV, Vu NB, Nguyen HT, et al. Characterization of the CD49f+/CD44+/CD24− single-cell derived stem cell population in basal-like DCIS cells. Oncotarget. 2016;7(39):63672–63684. doi:10.18632/oncotarget.10203

9. Andrews S. FastQC: a quality control tool for high throughput sequence data. Babraham Bioinformatics. 2010. http://www.bioinformatics.babraham.ac.uk/projects/fastqc

10. Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. Biostatistics. 2007;8(1):118-127. doi:10.1093/biostatistics/kxj037

11. Dobin A, Davis CA, Schlesinger F, et al. STAR: ultrafast universal RNA-seq aligner. Bioinformatics. 2013;29(1):15-21. doi:10.1093/bioinformatics/bts635

12. Engstrom PG, Steijger T, Sipos B, et al. Systematic evaluation of spliced alignment programs for RNA-seq data. Nat Methods. 2013;10(12):1185-1191. doi:10.1038/nmeth.2722

13. Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics. 2014;30(7):923-930. doi:10.1093/bioinformatics/btt656

14. Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol. 2014;15(12):550. doi:10.1186/s13059-014-0550-8

15. Anders S, Huber W. Differential expression analysis for sequence count data. Genome Biol. 2010;11(10):R106. doi:10.1186/gb-2010-11-10-r106

16. Zhu A, Ibrahim JG, Love MI. Heavy-tailed prior distributions for sequence count data: removing the noise and preserving large differences. Bioinformatics. 2019;35(12):2084-2092. doi:10.1093/bioinformatics/bty895

17. Subramanian A, Tamayo P, Mootha VK, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. Proc Natl Acad Sci USA. 2005;102(43):15545-15550. doi:10.1073/pnas.0506580102

18. Mootha VK, Lindgren CM, Eriksson KF, et al. PGC-1alpha-responsive genes involved in oxidative phosphorylation are coordinately downregulated in human diabetes. Nat Genet. 2003;34(3):267-273. doi:10.1038/ng1180

19. Nam D, Kim SY. Gene-set approach for expression pattern analysis. Brief Bioinform. 2008;9(3):189-197. doi:10.1093/bib/bbn001

20. Ben-Porath I, Thomson MW, Carey VJ, et al. An embryonic stem cell-like gene expression signature in poorly differentiated aggressive human tumors. Nat Genet. 2008;40(5):499-507. doi:10.1038/ng.127

21. Pece S, Tosoni D, Confalonieri S, et al. Biological and molecular heterogeneity of breast cancers correlates with their cancer stem cell content. Cell. 2010;140(1):62-73. doi:10.1016/j.cell.2009.12.007

22. Bhattacharya R, Bhattacharya S, Yin L, et al. DNA repair and therapeutic strategies in cancer stem cells. Cancers. 2023;15(6):1897. doi:10.3390/cancers15061897

23. Liberzon A, Birger C, Thorvaldsdottir H, Ghandi M, Mesirov JP, Tamayo P. The Molecular Signatures Database (MSigDB) hallmark gene set collection. Cell Syst. 2015;1(6):417-425. doi:10.1016/j.cels.2015.12.004

24. Kanehisa M, Goto S. KEGG: Kyoto Encyclopedia of Genes and Genomes. Nucleic Acids Res. 2000;28(1):27-30. doi:10.1093/nar/28.1.27

25. Jassal B, Matthews L, Viteri G, et al. The Reactome Pathway Knowledgebase. Nucleic Acids Res. 2020;48(D1):D498-D503. doi:10.1093/nar/gkz1031

26. Reimand J, Isserlin R, Voisin V, et al. Pathway enrichment analysis and visualization of omics data using g:Profiler, GSEA, Cytoscape and EnrichmentMap. Nat Protoc. 2019;14(2):482-517. doi:10.1038/s41596-018-0103-9

27. Smedley D, Haider S, Durinck S, et al. The BioMart community portal: an innovative alternative to large, centralized data repositories. Nucleic Acids Res. 2015;43(W1):W589-598. doi:10.1093/nar/gkv350

28. Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. CA Cancer J Clin. 2023;73(1):17-48. doi:10.3322/caac.21763

29. DeSantis CE, Ma J, Gaudet MM, et al. Breast cancer statistics, 2019. CA Cancer J Clin. 2019;69(6):438-451. doi:10.3322/caac.21583

30. Warner ET, Tamimi RM, Hughes ME, et al. Racial and ethnic differences in breast cancer survival: mediating effect of tumor characteristics and sociodemographic and treatment factors. J Clin Oncol. 2015;33(20):2254-2261. doi:10.1200/JCO.2014.57.1349

31. Islami F, Guerra CE, Minihan A, et al. American Cancer Society report on the status of cancer disparities in the United States, 2021. CA Cancer J Clin. 2022;72(2):112-143. doi:10.3322/caac.21703

32. Speer RM, Zhou X, Volk LB, Liu KJ, Hudson LG. Arsenic and cancer: evidence and mechanisms. Adv Pharmacol. 2023;96:151-202. doi:10.1016/bs.apha.2022.08.001

33. Zhou X, Speer RM, Volk L, Hudson LG, Liu KJ. Arsenic co-carcinogenesis: inhibition of DNA repair and interaction with zinc finger proteins. Semin Cancer Biol. 2021;76:86-98. doi:10.1016/j.semcancer.2021.05.009

34. Saintilnord WN, Fondufe-Mittendorf Y. Arsenic-induced epigenetic changes in cancer development. Semin Cancer Biol. 2021;76:195-205. doi:10.1016/j.semcancer.2021.03.019

35. Andrew AS, Karagas MR, Hamilton JW. Decreased DNA repair gene expression among individuals exposed to arsenic in United States drinking water. Int J Cancer. 2003;104(3):263-268. doi:10.1002/ijc.10968

36. Nollen M, Ebert F, Moser J, Mullenders LH, Hartwig A, Schwerdtle T. Impact of arsenic on nucleotide excision repair: XPC function, protein level, and gene expression. Mol Nutr Food Res. 2009;53(5):572-582. doi:10.1002/mnfr.200800480

37. Ding W, Liu W, Cooper KL, et al. Inhibition of poly(ADP-ribose) polymerase-1 by arsenite interferes with repair of oxidative DNA damage. J Biol Chem. 2009;284(11):6809-6817. doi:10.1074/jbc.M805566200

38. Piatek K, Schwerdtle T, Hartwig A, Bal W. Inorganic arsenic disrupts the zinc finger domain of the DNA repair protein XPA. Chem Res Toxicol. 2008;21(3):600-606. doi:10.1021/tx7003135

39. Hamadeh HK, Vargas M, Lee E, Menzel DB. Arsenic disrupts cellular levels of p53 and mdm2: a potential mechanism of carcinogenesis. Biochem Biophys Res Commun. 1999;263(2):446-449. doi:10.1006/bbrc.1999.1395

40. Groisman R, Polanowska J, Kuraoka I, et al. The ubiquitin ligase activity in the DDB2 and CSA complexes is differentially regulated by the COP9 signalosome in response to DNA damage. Cell. 2003;113(3):357-367. doi:10.1016/S0092-8674(03)00316-7

41. Ahmad A, Robinson AR, Duensing A, et al. ERCC1-XPF endonuclease facilitates DNA double-strand break repair. Mol Cell Biol. 2008;28(16):5082-5092. doi:10.1128/MCB.00293-08

42. Staresincic L, Fagbemi AF, Enzlin JH, et al. Coordination of dual incision and repair synthesis in human nucleotide excision repair. EMBO J. 2009;28(8):1111-1120. doi:10.1038/emboj.2009.49

43. Tsodikov OV, Ivanov D, Orelli B, et al. Structural basis for the recruitment of ERCC1-XPF to nucleotide excision repair complexes by XPA. EMBO J. 2007;26(22):4768-4776. doi:10.1038/sj.emboj.7601894

44. Bailey ZD, Krieger N, Agenor M, Graves J, Linos N, Bassett MT. Structural racism and health inequities in the USA: evidence and interventions. Lancet. 2017;389(10077):1453-1463. doi:10.1016/S0140-6736(17)30569-X

45. Kim J, Lee JJ, Kim J, Gardner D, Beachy PA. Arsenic antagonizes the Hedgehog pathway by preventing ciliary accumulation and reducing stability of the Gli2 transcriptional effector. Proc Natl Acad Sci USA. 2010;107(30):13432-13437. doi:10.1073/pnas.1006822107

46. Chang Y, Zhu X, Yao M, et al. Arsenic trioxide inhibits the growth of cancer stem cells derived from small cell lung cancer by downregulating stem cell-maintenance factors and inducing apoptosis via the Hedgehog signaling blockade. Transl Lung Cancer Res. 2020;9(4):1379-1393. doi:10.21037/tlcr-20-467

47. Doheny D, Manore SG, Wong GL, Lo HW. Canonical Hedgehog pathway and noncanonical GLI transcription factor activation in cancer. Cells. 2020;9(9):2114. doi:10.3390/cells9092114

48. Grivennikov SI, Greten FR, Karin M. Immunity, inflammation, and cancer. Cell. 2010;140(6):883-899. doi:10.1016/j.cell.2010.01.025

49. Karin M. Nuclear factor-kappaB in cancer development and progression. Nature. 2006;441(7092):431-436. doi:10.1038/nature04870

50. Reichard JF, Puga A. Effects of arsenic exposure on DNA methylation and epigenetic gene regulation. Epigenomics. 2010;2(1):87-104. doi:10.2217/epi.09.45